Growth, collisions and lamplighter structure of generic triangle and orthoscheme reflection groups

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Abstract

This paper consolidates a group of preprints on the reflection groups of generic Euclidean orthoschemes and their growth. The unconditional core is: the right-angled Coxeter envelope W_n has rational growth $(1+t)^{\lfloor n/2 \rfloor+1}/K_{n+1}(t)$ with rate $r_n = 1+2\cos(2\pi/(n+3))$; a dichotomy identifies the collision depth of a leg tuple with half the shortest kernel length; for $n \geq 3$ every length-six kernel element lies in a rank-two standard parabolic and occurs only at dihedral angle $\pi/3$; over $\mathbb{Q}^n_{>0}$ the rank-two kernel condition is decided by the positional statistics of the leg tuple, the Diophantine input being three rank-zero elliptic curves, and the corresponding statement over $\mathbb{R}$ is false. In the plane the kernel of the triangle representation is nontrivial and the growth series of the generic triangle group departs from the envelope at depth 11. The growth rate of the planar right-triangle group is $\beta_2 = 1/\sqrt{q^*} = 1.4916177871\ldots$, where q^* is the smallest positive zero of a Hahn–Exton q-cosine; $\beta_2 < 3/2$ is unconditional. The travel resolvent is transcendental over $\mathbb{Q}(x)$ unconditionally.

The material on embeddings is new and is stated first. Section 6 proves that an irreducible, infinite, non-affine Coxeter system with a symmetric Cartan matrix of rank $|S|-1$ has a faithful $(|S|-1)$-dimensional orthogonal representation, and that for right-angled systems this can be transported to the compact group $O(|S|-1)$; in particular W_n embeds in $O(n)$, and hence in $\mathrm{Isom}(\mathbb{R}^n)$, for every $n \geq 3$, while $n = 2$ is a genuine exception. Section 3 contains the other statement not in the source preprints: for every $n \geq 2$, any embedding of W_n into $\mathrm{Isom}(\mathbb{R}^n)$ has injective linear part, so the ambient embedding problem is equivalent to the question of whether W_n embeds in the *compact* group $O(n)$. At $n = 2$ this recovers the planar obstruction from solvability of $O(2)$ alone, without amenability of $\mathrm{Isom}(\mathbb{R}^2)$, and it isolates non-commutativity of $O(n)$ as the exact point at which the planar argument fails for $n \geq 3$.

Section 1 states, for each result, what it is conditional on. Three hypotheses are open and are named wherever they are used: the strand-walk transfer model (M), the junction pairing $(R-J)$, and the invariance of the travel block (T). Transcendence of the full growth series U and of its relaxed companion V is conditional on these and is not claimed here. Irrationality of β_2 is open.

Sections 4 and 5 settle the three-dimensional case. The conjecture that the orthoscheme representation is faithful at every Class C rational leg tuple is false, with explicit counterexamples, the shortest kernel element found having length 76. On the other hand the tuple $(1, 2, 11)$ is faithful, proved by identifying W_3 as $F_2 \rtimes (\mathbb{Z}/2\times\mathbb{Z}/2)$ and recognising the image of the free factor as the classical free group of rotations by $\arccos(3/5)$ about perpendicular axes. Consequently, in dimension three, $\bigcap_a \ker \rho_a$ is trivial, the generic growth series is the rational envelope in every degree, W_3 embeds in $\mathrm{Isom}(\mathbb{R}^3)$, and the generic 3-orthoscheme group is finitely presented.

1 What is new, what is known, and what is conditional

1.1 Prior art

The following are not ours and are used as such. Isola and Piergallini proved that the generic triangle reflection group has free metabelian rotation subgroup of rank two and is not finitely presented. Droms, Lewin and Servatius, and Myasnikov, Roman'kov, Ushakov and Vershik, gave the length formula $\|\phi\|_1 + 2\,\mathrm{st}(\phi)$ for free solvable quotients F/N'; Sale gave the metric behaviour of the Magnus embedding. Steinberg's formula for Poincaré series of Coxeter groups is standard, as are the independence polynomials of paths and the location of their roots. Adamczewski, Dreyfus and Hardouin proved hypertranscendence of the solutions of the relevant q-difference equation for all q not a root of unity, which is stronger than the SL_2 statement recorded in §10; our increment there is only the removal of a transcendence hypothesis on q in that one special case.

1.2 Claim labels

Every numbered statement below carries exactly one of PROVED (complete human proof), VERIFIED (formalized in Lean 4, axioms audited), HEURISTIC (computational support only), or CONJECTURE. The following are VERIFIED, in `lean/with_mathlib/TranslationTrick.lean` and `lean/with_mathlib/TransferChain.lean`, each cold-elaborated at exit 0 with standard axioms only and no `sorry`: Proposition 4.1 and its contrapositive; det = 1 for the travel and bulk transfer matrices; nilpotence of their increments; and the half-shift identity. The effective strength of a result is the weakest label in its dependency chain.

1.3 The three open hypotheses

Hypothesis 1.1 (transfer model, (M))**.** The strand-walk model computes the relaxed word length, that is, the crossing optimisation of the site-cost chain is the relaxed length.

Hypothesis 1.2 (junction pairing, $(R{-}J)$)**.** The junction pairing $\Pi_y(q_m)$ is nonzero at infinitely many travel poles q_m.

Hypothesis 1.3 (travel-block invariance, (T))**.** The denominator $1 - \Sigma_1$ is the identical, cycle-marker-free object in the generating functions of U and of V, so U and V share the travel poles.

Hypothesis 1.3 is HEURISTIC. Its strand-walk argument rests on the localisation of cycles, checked exhaustively on the 3869 pure-travel elements of true length at most 16, and on the metric identity $\ell_T = \ell_R + 2c$, whose upper bound is PROVED and whose lower bound is open. Hypothesis 1.2 is CONJECTURE. Hypothesis 1.1 is HEURISTIC and everything in the block-assembly sections is conditional on it.

Remark 1.4 (what is not claimed). Three claims that appeared in a draft consolidation are withdrawn here because they are not supported by the sources.

(i) Irrationality of β_2 is *open*. It is not proved, and no effective irrationality measure for β_2 is available. What is proved is $\beta_2 \neq 3/2$, by the squeeze of §8. The reformulation known is that β_2 is irrational if and only if a Hahn–Exton function omits the value 0 at the relevant rational points; that is a restatement, not a proof, and it supplies no exponent.

(ii) Hypothesis 1.3 is *not* a theorem.

(iii) The lower bound of the metric identity $\ell_T = \ell_R + 2c$ is *open*; only the upper bound is PROVED. Statements that consume the identity inherit that standing.

2 The envelope and the dichotomy

Let $O(a_1, \ldots, a_n) \subset \mathbb{R}^n$ be the right-corner orthoscheme with vertices $V_0 = 0$ and $V_k = V_{k-1} + a_k e_k$, and let $\widehat{R}_0, \ldots, \widehat{R}_n$ be the reflections in its facets, generating $W(a) \leq \mathrm{Isom}(\mathbb{R}^n)$. Facets i, j with $|i - j| \geq 2$ are perpendicular for every leg tuple, so with Γ_n the graph on $\{0, \ldots, n\}$ having an edge whenever $|i - j| \geq 2$,

$$W_n = \big\langle R_0, \ldots, R_n \;\big|\; R_i^2 = 1,\ (R_i R_j)^2 = 1\ (|i - j| \geq 2) \big\rangle \tag{1}$$

is the right-angled Coxeter group on Γ_n, and $\rho_a \colon W_n \twoheadrightarrow W(a)$.

Theorem 2.1 (Envelope; PROVED)**.** *$W_n(t) = (1 + t)^{\lfloor n/2 \rfloor + 1} / K_{n+1}(t)$, with $K_0 = K_1 = 1$, $K_{2j} = K_{2j-1} - tK_{2j-2}$ and $K_{2j+1} = (1 + t)K_{2j} - tK_{2j-1}$. The growth rate is $r_n = 1 + 2\cos(2\pi/(n + 3))$.*

Theorem 2.2 (Dichotomy; PROVED)**.** *For a marked quotient $Q = G/N$ with $K(N) = \min_{1 \neq g \in N} \ell_G(g)$ and $h = \lceil K/2 \rceil$, the sphere sizes satisfy $s_d(Q) = s_d(G)$ for $d < h$ and $s_h(Q) < s_h(G)$. Hence $\mathrm{cd}_n(a) = K(\ker \rho_a)/2$.*

Theorem 2.3 (Length six; PROVED)**.** *For $n \geq 3$ and $a \in \mathbb{R}^n_{>0}$, if $w \in \ker \rho_a$ has $\ell_{W_n}(w) = 6$ then $w = (R_i R_{i+1})^{\pm 3}$ for some i with $\theta_{i,i+1} = \pi/3$.*

Theorem 2.4 (Rank-two classification; PROVED)**.** *For $a \in \mathbb{Q}^n_{>0}$,* $\ker \rho_a$ *meets a rank-two standard parabolic nontrivially if and only if $e(a) + t(a) \geq 1$, where e counts endpoint equal pairs and t counts three consecutive equal legs. The Diophantine input is the triviality of the rational points on three quartics with Jacobians 24a1, 72a2 and $y^2 = x^3 - x$, all of rank zero. Over $\mathbb{R}$ the statement is false: $a = (\sqrt{3}, 1, 5)$ has a rank-two kernel element with no equal legs.*

3 The ambient embedding reduces to a compact group

This section is the one place where the present paper goes beyond the sources. Write $L\colon\ \mathrm{Isom}(\mathbb{R}^n) \to O(n)$ for the linear part, with kernel the translation group $\mathbb{R}^n$.

Lemma 3.1 (PROVED)**.** *For every $n \geq 2$, W_n has no nontrivial normal abelian subgroup.*

Proof. The Coxeter diagram of the system (1) has an edge exactly where $m_{ij} \geq 3$, that is, exactly where $\{i, j\}$ is a non-edge of Γ_n, that is, exactly where $|i - j| = 1$. So the diagram is the path on $n + 1$ nodes with every edge labelled ∞. The path is connected, so (W_n, S) is irreducible. Every label is ∞, so W_n is infinite and is not of affine type, affine Coxeter systems having all labels finite.

Let $A \trianglelefteq W_n$ be abelian; A is amenable, so it lies in the amenable radical of W_n. We show that radical is trivial, in two steps.

First it is finite. An irreducible, infinite, non-affine Coxeter group contains a rank-one isometry for its action on the Davis complex (Caprace–Fujiwara), and a group acting properly cocompactly on a CAT(0) space with a rank-one isometry, and which is not virtually cyclic, is acylindrically hyperbolic. W_n is not virtually cyclic: for $n \geq 3$ it contains $\langle R_0, R_2 \rangle \times \langle R_1 \rangle$, and for $n = 2$ it is $(\mathbb{Z}/2 \times \mathbb{Z}/2) * \mathbb{Z}/2$, which contains a free group of rank two. For an acylindrically hyperbolic group the amenable radical coincides with the finite radical, the unique maximal finite normal subgroup (Osin). Note that acylindrical hyperbolicity gives finiteness here, not triviality; the second step is needed.

Second it is trivial. The finite radical of a Coxeter group is the direct product of its finite irreducible components, and W_n is irreducible and infinite, so it has no finite component and its finite radical is trivial. Hence $A = 1$. ∎

Proposition 3.2 (Reduction to $O(n)$; PROVED)**.** *Let $n \geq 2$ and let $\varphi\colon W_n \to \mathrm{Isom}(\mathbb{R}^n)$ be an injective homomorphism. Then $L \circ \varphi$ is injective. Consequently*

$$W_n \text{ embeds in } \mathrm{Isom}(\mathbb{R}^n) \quad\Longleftrightarrow\quad W_n \text{ embeds in } O(n).$$

Proof. $\ker(L \circ \varphi) = \varphi^{-1}\big(\varphi(W_n) \cap \mathbb{R}^n\big)$ is normal in W_n and is isomorphic to a subgroup of the abelian group $\mathbb{R}^n$, hence abelian. By Lemma 3.1 it is trivial. The equivalence follows, one direction being $O(n) \leq \mathrm{Isom}(\mathbb{R}^n)$. □

Corollary 3.3 (The plane, without amenability; PROVED)**.** *W_2 does not embed in* $\mathrm{Isom}(\mathbb{R}^2)$.

Proof. $O(2) = SO(2) \rtimes \mathbb{Z}/2$ is metabelian, hence solvable, hence contains no free subgroup of rank two. $W_2 = (\mathbb{Z}/2 \times \mathbb{Z}/2) * \mathbb{Z}/2$ is a free product of finite groups of orders 4 and 2 and is not infinite dihedral, so it contains a free subgroup of rank two. So W_2 does not embed in $O(2)$, and Proposition 3.2 transfers this to $\mathrm{Isom}(\mathbb{R}^2)$. □

Remark 3.4 (where the planar argument really fails)*.* The published planar obstruction runs through amenability of $\mathrm{Isom}(\mathbb{R}^2)$. That route is unavailable for $n \geq 3$ because $SO(3)$ contains a free subgroup of rank two, so $\mathrm{Isom}(\mathbb{R}^n)$ is not amenable as an abstract group. Proposition 3.2 relocates the obstruction: the translation part is never the issue in any dimension, and the planar proof uses exactly one property of the ambient group, namely that $O(2)$ is solvable. For $n \geq 3$ the target $O(n)$ is a compact, non-solvable Lie group, and that single change is the whole of the difficulty.

Remark 3.5 (what the reduction buys)*.* Replacing $\mathrm{Isom}(\mathbb{R}^n)$ by $O(n)$ replaces a non-compact target by a compact one, and compactness is usable. If $W_n \leq O(n)$, its closure H is a compact Lie subgroup of $O(n)$, W_n is dense in H, and H is infinite since W_n is. Hence H° is a connected compact Lie group of dimension at most $n(n-1)/2$, the subgroup $W_n \cap H^\circ$ has finite index in W_n and is dense in H°. In particular an embedding would give W_n a faithful orthogonal representation of degree n. This is a much more rigid demand than a faithful action by isometries of $\mathbb{R}^n$, and it is the form in which we propose Problem 11.15 be attacked.

Remark 3.6 (the point group, measured)*.* Proposition 3.2 says the linear part carries everything, so the object to measure is the point group $\pi \circ \rho_a\colon W_n \to O(n)$, not ρ_a. If the point group is injective then so is ρ_a, so a faithful point group would settle Problems 11.13, 11.15 and 11.16 together; the converse fails, and a deviation of the point group is not a deviation of ρ_a. The point group has not previously been measured on its own. Running it against the envelope at two independent primes $2^{61} - 1$ and $2^{62} - 57$, at Class C leg tuples, gives agreement to the following depths:

n	2	3	4	5	6
legs	$(1,2)$	$(1,2,3)$	$(1,2,3,5)$	$(3,1,4,2,5)$	$(1,2,3,5,7,4)$
agrees to depth	**2**	22	17	15	13

and a second tuple at $n = 3$, namely $(2,5,3)$, and at $n = 4$, namely $(3,1,4,2)$, give the same depths as the first. The $n = 2$ column is a negative control and it fires: there the point group lies in $O(2)$, which is solvable, while W_2 contains a free group of rank two, so a deviation is forced, and it occurs at depth 3, the earliest depth not already fixed by the $(R_0R_2)^2 = 1$ relation. For $n \geq 3$ no deviation is found. This is HEURISTIC and decides nothing: it is consistent both with faithfulness and with a horizon beyond the computed range. What it does establish, at each tuple listed, is that the point group is injective on the ball of the stated radius, hence so is ρ_a; at $n = 3$ that improves the previously recorded depth for ρ_a from 19 to 22, and does so for the stronger object.

4 Class C faithfulness is false

Remark 3.6 found Class C tuples at which the point group is *not* injective. That is not by itself a statement about ρ_a, the point group being a quotient. The following elementary observation converts it into one, and needs neither Lemma 3.1 nor any CAT(0) input.

Proposition 4.1 (Translations commute; PROVED). *Let $a \in \mathbb{R}^n_{>0}$ and let $u \in W_n$ satisfy $\pi(\rho_a(u)) = I$, that is, $\rho_a(u)$ has trivial linear part. Then for every $g \in W_n$,*

$$[\,u,\ gug^{-1}\,] \ \in\ \ker \rho_a.$$

Proof. An isometry of $\mathbb{R}^n$ with trivial linear part is a translation, so $\rho_a(u)$ is a translation. For any g, $\rho_a(gug^{-1}) = \rho_a(g)\rho_a(u)\rho_a(g)^{-1}$ is a conjugate of a translation by an isometry, hence again a translation. Two translations of $\mathbb{R}^n$ commute, so the commutator of their preimages is killed. □

Theorem 4.2 (Conjecture C is false; PROVED). *There are Class C tuples, that is with pairwise distinct coordinates, at which ρ_a is not injective, in dimensions three and four. In $\mathbb{Q}^3_{>0}$ this holds at*

$$(1,3,2),\quad (2,3,9),\quad (2,6,9),\quad (3,1,5),\quad (3,2,6),\quad (4,9,7),\quad (6,11,7)$$

and at their reversals, where $a = (3,1,5)$ gives an element of $\ker \rho_a$ of length 76; and in $\mathbb{Q}^4_{>0}$ it holds at $(1,2,3,4)$ and $(2,1,3,4)$ and their reversals, each giving an element of length 86. These are all the primitive pairwise-distinct tuples with entries at most 12 respectively 5 at which the point group collapses.

Proof. Take $a = (3,1,5)$; the other tuples are identical in form. Breadth-first search in W_3, using the right-angled normal form and exact rational arithmetic for the linear reflections, produces two elements $w_1 \neq w_2$ of W_3 of length 11 with the same linear part; the sphere count of the point group at depth 11 falls short of the envelope, at each of two independent primes. Put $u = w_1 w_2^{-1}$, of length 22. Then $\pi(\rho_a(u)) = I$ by construction, and $u \neq 1$ in W_3, verified in the Tits geometric representation of the Coxeter system, which is faithful for every Coxeter system and has integer matrices here, so the verification is exact. By Proposition 4.1, $c = [u,\ R_1 u R_1]$ lies in $\ker \rho_a$, and $c \neq 1$ in W_3, again by the Tits representation; reducing c gives a geodesic of length 76. Finally the coordinates of a are pairwise distinct, so a is Class C. The dimension-four tuples are identical in form, the only change being that W_4 replaces W_3 throughout; nothing in the argument is special to dimension three, since Proposition 4.1 holds for all n. □

Remark 4.3 (why this was not seen). The counterexamples are invisible to a direct search. By the dichotomy the deviation of ρ_a sits at depth $\lceil K/2 \rceil$, and here $K \leq 76$, so the first depth at which the orbit growth of $W(a)$ departs from the envelope is somewhere below 38 but far above any reachable breadth-first search: the envelope sphere at depth 38 is $9 \cdot 2^{36}$, already above $6 \cdot 10^{11}$. A direct computation on ρ_a at these tuples sees agreement with the envelope for as long as it can run, and indeed does. What makes the counterexamples accessible is that the *linear part* fails much earlier, at depth 11 to 13, where the search is trivial. That is the practical content of Proposition 3.2: the obstruction lives in $O(n)$, and it can be seen there.

Corollary 4.4 (a necessary condition; PROVED). *If ρ_a is injective then the point group $\pi \circ \rho_a$ is injective. Equivalently, a point-group collision at a certifies that ρ_a is not injective.*

Proof. $\ker(\pi \circ \rho_a)$ is normal in W_n, and if ρ_a is injective it is isomorphic to its image, a subgroup of the translation group, hence abelian. By Lemma 3.1 it is trivial. □

Theorem 4.5 (the three problems have one answer; PROVED)**.** *Fix $n \geq 3$ and let*

$$U_n \;=\; \bigcap_a \ker\bigl(\pi \circ \rho_a\bigr)$$

be the universal kernel of the point group, the set of $w \in W_n$ whose image has trivial linear part at every *leg tuple.*

(i) *If $U_n = 1$ then ρ_a is injective for every a outside a countable union of proper algebraic subsets of the positive orthant, hence for a co-null set of tuples. Consequently $N_{\mathrm{gen}} = 1$; W_n embeds in $O(n)$ and so in $\mathrm{Isom}(\mathbb{R}^n)$; and the generic n-orthoscheme group is W_n, which is finitely presented.*

(ii) *If $U_n \neq 1$, pick $1 \neq u \in U_n$. If $[u, gug^{-1}] \neq 1$ in W_n for some g, then that commutator lies in $\ker \rho_a$ for* every *a, so $N_{\mathrm{gen}} \neq 1$ and the generic group is a proper quotient of W_n.*

So $U_n = 1$ answers all three affirmatively at once. The converse is one-directional, and only for the first: $U_n \neq 1$ refutes Problem 11.13, but it refutes neither Problem 11.15 nor Problem 11.16, since a proper quotient of W_n may still be finitely presented, and W_n may still embed in $O(n)$ by a map that is not the point group of any orthoscheme.

Proof. (i) Fix $w \neq 1$. The entries of $\pi(\rho_a(w))$ are rational functions of a whose denominators are sums of squares of the legs, nonvanishing on the positive orthant, so $Z_w = \{a : \pi(\rho_a(w)) = I\}$ is closed and algebraic there; it is proper because $U_n = 1$ supplies a tuple outside it. W_n is countable, so $\bigcup_{w\neq 1} Z_w$ is a countable union of proper algebraic subsets and is null. For a outside it, $\pi \circ \rho_a$ is injective, hence so is ρ_a. Injectivity of $\pi \circ \rho_a$ at a single such a realises W_n inside $O(n)$, and $O(n) \leq \mathrm{Isom}(\mathbb{R}^n)$. Finally $N_{\mathrm{gen}} \subseteq \ker \rho_a = 1$, so the generic group is W_n, a right-angled Coxeter group and therefore finitely presented.

(ii) For $u \in U_n$ the image $\rho_a(u)$ has trivial linear part at every a, so it is a translation at every a. Proposition 4.1 then places $[u, gug^{-1}]$ in $\ker \rho_a$ for every a and every g, whence it lies in N_{gen}. □

Remark 4.6 (what this changes). In dimension three $U_3 = 1$, since Theorem 5.2 exhibits a faithful tuple, and (i) recovers all three closures at once. In dimension four the families of §4 show that many individual tuples lie in some Z_w; they say nothing about U_4, because each of those Z_w is a proper hypersurface and not the whole orthant. Indeed the exhaustive scan is evidence for $U_4 = 1$: were U_4 nontrivial, some word would fail at *every* tuple, and no tuple in the scan fails at all outside the two loci.

What (i) needs is therefore not a faithful tuple in dimension four but the triviality of U_4, and the two are equivalent. We do not prove it, but the computations bound it from below.

It should be stressed that Problems 11.15 and 11.16 require strictly less. The embedding problem asks only for *some* injective homomorphism $W_n \to O(n)$, which need not be a point group of an orthoscheme and need not be by reflections at all; and finite presentation of W_n/N_{gen} does not require $N_{\mathrm{gen}} = 1$. So a negative answer for U_n would leave both of them open, and either could be settled affirmatively by a construction that says nothing about U_n.

Proposition 4.7 (a lower bound on the universal kernel; PROVED)**.** *U_n contains no nontrivial element of length at most $2d_n$, where*

n	3	4	5	6
d_n	22	18	15	14
no element of length $\leq$	44	36	30	28

Proof. Fix n and a leg tuple at which the point group has been computed. Since $U_n \subseteq \ker(\pi \circ \rho_a)$, any element of U_n is an element of that kernel. By the dichotomy, Theorem 2.2, if the orbit growth of the image agrees with the envelope through depth d then the shortest nontrivial kernel element has length at least $2d+1$. The tabulated d_n are the depths to which agreement is verified, at two independent primes, at the tuples $(1,2,11)$, $(1,2,3,5)$, $(3,1,4,2,5)$ and $(1,2,3,5,7,4)$ respectively; the value at $n=4$ is confirmed at the second tuple $(3,1,4,2)$ as well. Reduction modulo a prime can only identify elements, so a computed sphere count is a lower bound for the characteristic-zero count, while the envelope is an upper bound; agreement therefore certifies the count exactly. $\square$

Remark 4.8 (a degeneration constraint on U_n, and why it is not enough). Sending a single interior leg to infinity degenerates exactly one consecutive pair. With $a_i \to \infty$ the cosine $\cos(m_{i-1}, m_i) \to 0$ while every other consecutive cosine tends to a value strictly between -1 and 0, and no two normals become parallel. The Gram matrix therefore becomes block diagonal and the limiting point group is a direct product of two smaller configurations of the same shape.

Since the entries of $\pi(\rho_a(w))$ are continuous on the closure, an element of U_n must die in every such limit, so U_n is contained in the normal closure of $[R_{i-1}, R_i]$ for each interior i, hence in the intersection of those normal closures.

That intersection is not trivial, so the constraint cannot by itself give $U_n = 1$. Equivalently the map $W_n \to \prod_i W_{\Gamma_n + \{i,i+1\}}$, adding one consecutive edge at a time, is not injective: already at $n=3$ the image sphere at depth 6 has 141 elements against 144 in W_3. The obstruction to promoting the constraint by induction is visible in the same picture, since a split at an interior pair leaves a block of two vectors, whose own universal kernel is nontrivial for the reason that $O(2)$ is solvable, so the induction has no base.

Remark 4.9 (what the bound does and does not say). At $n=3$ the bound is superseded: $U_3 = 1$ outright. For $n \geq 4$ it is the only unconditional information we have about U_n, and it is not close to triviality, since U_n could still be generated in some length beyond the computed range. The bound degrades with n because the envelope grows faster, so a fixed memory budget reaches a smaller depth; it is a limitation of the search and not evidence that U_n is more likely to be nontrivial in higher dimensions.

Remark 4.10 (what remains open). Theorem 4.2 settles the arithmetic form of Problem 11.13 negatively. It does *not* settle the generic form: the tuples above are single points, and $N_{\text{gen}} = \bigcap_a \ker \rho_a$ may still be trivial, since a kernel element at one tuple need not be a kernel element at another. It also leaves Problem 11.15 open, an abstract embedding not being required to come from an orthoscheme.

5 A faithful tuple in dimension three

Lemma 5.1 (Structure of W_3; PROVED)**.** *Put $A = R_0R_1$, $C = R_2R_3$ and $V = \langle R_0, R_3 \rangle$. Then $V \cong \mathbb{Z}/2 \times \mathbb{Z}/2$, $\langle A, C \rangle$ is free of rank two with basis $\{A, C\}$, and $W_3 = \langle A, C \rangle \rtimes V$, the action being $R_0AR_0 = A^{-1}$, $R_0CR_0 = C$, $R_3AR_3 = A$, $R_3CR_3 = C^{-1}$.*

Proof. The four conjugation identities are immediate from the defining relations, using that R_0 commutes with R_2, R_3 and R_1 with R_3. Hence V normalises $\langle A, C \rangle$, and since $R_1 = R_0A$ and $R_2 = CR_3$ the product $\langle A, C \rangle V$ is all of W_3.

Let $\varphi \colon W_3 \to (\mathbb{Z}/2)^2$ send $R_0, R_1 \mapsto (1,0)$ and $R_2, R_3 \mapsto (0,1)$; the defining relations are respected, so φ is well defined, and it is onto. Every nontrivial element of a finite parabolic $\langle R_i, R_j \rangle$ has nonzero image, and every torsion element of a right-angled Coxeter group is conjugate into a finite parabolic, so $\ker \varphi$ is torsion free. As Γ_3 is the path on four vertices, W_3 is the fundamental group of a graph of groups with three vertex groups $\mathbb{Z}/2 \times \mathbb{Z}/2$ and two edge groups $\mathbb{Z}/2$, so it is virtually free with $\chi(W_3) = 3/4 - 2/2 = -1/4$. A torsion-free virtually free group is free, so $\ker\varphi$ is free, of index 4 and hence of Euler characteristic -1, that is of rank two. Finally $\langle A, C\rangle \subseteq \ker\varphi$, $\langle A, C\rangle \cap V = 1$ because φ is injective on V, and $\langle A, C\rangle V = W_3$; so $\langle A, C\rangle$ has index 4 and equals $\ker\varphi$. A free group of rank two generated by two elements has them as a basis. □

Theorem 5.2 (A faithful tuple; PROVED). *At $a = (1, 2, 11)$ the point group $\pi \circ \rho_a : W_3 \to O(3)$ is injective. Hence ρ_a is injective.*

Proof. Write α for the angle with $\cos\alpha = 3/5$, $\sin\alpha = 4/5$. At this tuple $|m_1|^2 = 5$ and $|m_2|^2 = 125$, and a direct computation gives

$$\pi(\rho_a(A)) = \begin{pmatrix} 3/5 & -4/5 & 0 \\ 4/5 & 3/5 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \pi(\rho_a(C)) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -117/125 & -44/125 \\ 0 & 44/125 & 117/125 \end{pmatrix},$$

the first being the rotation about e_3 by α and the second the rotation about e_1 by 3α, since $\cos 3\alpha = 4\cos^3\alpha - 3\cos\alpha = -117/125$ and $\sin 3\alpha = 3\sin\alpha - 4\sin^3\alpha = 44/125$. So with B the rotation about e_1 by α, which is not itself in the group, $\pi(\rho_a(C)) = B^3$.

The rotations by α about two perpendicular axes generate a free group of rank two. This is the classical free-rotation theorem in exactly the form stated for $\cos\alpha = 3/5$ (Wagon, *The Banach–Tarski Paradox*, Thm. 2.1), proved by the 5-adic argument: for a reduced word w of length n whose first applied letter is $A^{\pm 1}$, the vector $5^n\, w\,(1, 0, 0)^T$ is integral and its second coordinate is not divisible by 5, so $w \neq I$; a word not of that form is conjugate to one that is, or is a power of B, which has infinite order because $3/5$ is not the cosine of a rational multiple of π (Niven). The invariant was checked independently over all 4,782,968 reduced words of length at most 14 of that form, in exact integer arithmetic. A subgroup of a free group is free, and A and B^3 form a basis of a free group of rank two, since a reduced word in A and B^3 substitutes to a word alternating A-blocks with B-blocks of nonzero exponent, which is already reduced and nonempty. Hence $\langle \pi\rho_a(A), \pi\rho_a(C)\rangle$ is free of rank two on those two elements, so by Lemma 5.1 the restriction of $\pi \circ \rho_a$ to $\langle A, C\rangle$ is injective.

The image of V consists of the four distinct diagonal matrices I, $\mathrm{diag}(-1, 1, 1)$, $\mathrm{diag}(1, 1, -1)$, $\mathrm{diag}(-1, 1, -1)$, so V injects. The image of $\langle A, C\rangle$ is free, hence torsion free, while every nontrivial element of the image of V is an involution, so the two images intersect trivially. Now if $\pi\rho_a(kv) = 1$ with $k \in \langle A, C\rangle$ and $v \in V$, then $\pi\rho_a(v) = \pi\rho_a(k)^{-1}$ lies in that intersection, so $\pi\rho_a(v) = 1$ and $\pi\rho_a(k) = 1$, whence $v = 1$ and $k = 1$. So $\pi \circ \rho_a$ is injective, and $\ker\rho_a \subseteq \ker(\pi \circ \rho_a) = 1$. □

Corollary 5.3 (Three consequences in dimension three; PROVED). *For $n = 3$:*

(i) $N_{\mathrm{gen}} = \bigcap_a \ker\rho_a = 1$, *so ρ_a is injective for every a in a co-null set of leg tuples, and the generic 3-orthoscheme growth series is the rational envelope $W_3(t) = (1+t)^2/(1-2t)$ in every degree;*

(ii) *W_3 embeds in $O(3)$, hence in $\mathrm{Isom}(\mathbb{R}^3)$, answering the ambient embedding question affirmatively;*

(iii) *the generic 3-orthoscheme reflection group $W_3/N_{\mathrm{gen}} = W_3$ is finitely presented.*

Proof. (i) $N_{\mathrm{gen}} = \bigcap_a \ker\rho_a \subseteq \ker\rho_{(1,2,11)} = 1$ by Theorem 5.2. For the co-null set, fix $w \neq 1$ in W_3. The entries of $\rho_a(w)$ are rational functions of a whose denominators do not vanish on the positive orthant, so $Z_w = \{a \in \mathbb{R}^3_{>0} : \rho_a(w) = 1\}$ is closed and algebraic there, and proper because $(1,2,11) \notin Z_w$. A proper algebraic subset is null and W_3 is countable, so $\bigcup_{w\neq 1} Z_w$ is null; off it every ρ_a is injective and $u_d(a) = [t^d]W_3(t)$ for every d. (ii) The image $\pi(\rho_a(W_3)) \leq O(3)$ is isomorphic to W_3 by Theorem 5.2. (iii) W_3 is a right-angled Coxeter group, so it is finitely presented. □

Remark 5.4 (a second route to Theorem 4.2). Lemma 5.1 is tuple independent, so it yields a proof of non-faithfulness that does not use Proposition 4.1. Since $W_3 = \langle A, C\rangle \rtimes V$ with $\langle A, C\rangle$ free on $\{A, C\}$, and V always injects, the point group is injective if and only if the images of A and C generate a free group on those two elements. At $(1,3,2)$ and at $(3,1,5)$ a search over reduced words in $A^{\pm 1}, C^{\pm 1}$ in exact rationals finds the first collision at word length 7, and, writing c for C^{-1} and a for A^{-1},

$$AAcAccAAACACCA \quad \text{at } (1,3,2), \qquad AAcaaCAAACaacA \quad \text{at } (3,1,5)$$

are reduced words of length 14 whose images are the identity while they are nontrivial in W_3.

Theorem 5.5 (an infinite Class C family in dimension four; PROVED)**.** *Let $a = (a_1, a_2, a_3, a_4) \in \mathbb{Q}^4_{>0}$ have pairwise distinct coordinates and satisfy*

$$a_1a_3 = a_2a_4.$$

Then ρ_a is not injective. An element of $\ker \rho_a$ *of length* 54 *is exhibited, uniformly across the family.*

Proof. Take the two fixed words of length eight

$$w_1 = R_1R_2R_1R_0R_3R_2R_1R_4, \qquad w_2 = R_0R_3R_2R_1R_4R_3R_2R_3,$$

and put $u = w_1w_2^{-1}$, of length 16. These do not depend on a.

Computing $\pi(\rho_a(w_1)) - \pi(\rho_a(w_2))$ symbolically in $\mathbb{Q}(a_1, a_2, a_3, a_4)$, every one of its fourteen nonzero entries carries the factor

$$(a_1a_3 - a_2a_4)(a_1a_3 + a_2a_4).$$

The coordinates are positive, so the second factor never vanishes, and therefore $\pi(\rho_a(u)) = I$ if and only if $a_1a_3 = a_2a_4$. Both directions are confirmed directly: substituting $a_4 = a_1a_3/a_2$ makes the difference identically zero, while at $(1,2,5,7)$, off the locus, it is nonzero.

So on the locus $\rho_a(u)$ is a translation, and $u \neq 1$ in W_4 by the Tits representation. Proposition 4.1 places $c = [u,\ R_0uR_0]$ in $\ker \rho_a$, and $c \neq 1$ in W_4, again by the Tits representation; reducing c gives a geodesic of length 54. Since w_1, w_2 and hence c are independent of a, the same element works for every member of the family. □

Remark 5.6 (the family is genuinely infinite, and the criterion is sharp). The equation $a_1a_3 = a_2a_4$ has infinitely many pairwise-distinct positive integer solutions, for instance $(k,\ km,\ kmn,\ kn)$ for distinct k, m, n scaled to be primitive, so Theorem 5.5 exhibits infinitely many Class C tuples at which ρ_a fails. This is a stronger statement than the dimension-three refutation, which produced only a finite list.

The criterion was found by scanning the 828 primitive pairwise-distinct tuples with entries at most 8, of which 24 fail by depth 12; the sixteen that fail at depth exactly 8 are precisely those with $a_1a_3 = a_2a_4$. It was then frozen and tested out of sample on ten tuples outside that box, all of which deviate at depth 8, against four controls with $a_1a_3 \neq a_2a_4$, none of which deviates by depth 12. The proof above supersedes that evidence: the colliding pair is a single pair of words, the same for every tuple, and the locus on which its linear parts agree is cut out exactly by $a_1a_3 = a_2a_4$. The remaining failures, at depth 12, form a second family, characterised in Theorem 5.7.

Theorem 5.7 (the second dimension-four family; PROVED)**.** *Let $a \in \mathbb{Q}^4_{>0}$ have pairwise distinct coordinates and satisfy*

$$a_4 \left|a_1^2 - a_2^2\right| \;=\; 2\, a_1 a_2 a_3.$$

Then ρ_a is not injective, with a kernel element obtained from a collision at depth 12*.*

Proof. Take the fixed words $w_1 = R_1R_0R_1R_2R_1R_0R_1R_0R_3R_2R_1R_0$ and $w_2 = R_0R_3R_2R_1R_0R_1R_4R_3R_2R_1R_3R_4$ of length twelve, independent of a, and put $u = w_1 w_2^{-1}$. Each of the fifteen nonzero entries of $\pi(\rho_a(w_1)) - \pi(\rho_a(w_2))$, computed symbolically in $\mathbb{Q}(a_1, a_2, a_3, a_4)$, carries the factor

$$\bigl(a_1^2 a_4 - 2a_1a_2a_3 - a_2^2 a_4\bigr)\bigl(a_1^2 a_4 + 2a_1a_2a_3 - a_2^2 a_4\bigr),$$

together with monomials in the a_i, which are nonzero on the positive orthant. The two factors vanish respectively where $a_4(a_1^2 - a_2^2) = 2a_1a_2a_3$, requiring $a_1 > a_2$, and where $a_4(a_2^2 - a_1^2) = 2a_1a_2a_3$, requiring $a_2 > a_1$; together they are the displayed condition. On that locus $\rho_a(u)$ is a translation, $u \neq 1$ in W_4 by the Tits representation, and Proposition 4.1 supplies the kernel element as before. ☐

Theorem 5.8 (the third dimension-four family; PROVED)**.** *Let $a \in \mathbb{Q}^4_{>0}$ have pairwise distinct coordinates and satisfy*

$$a_1 \left|a_3^2 - a_4^2\right| \;=\; 2\, a_2 a_3 a_4.$$

Then ρ_a is not injective, with a kernel element from a collision at depth 12*.*

Proof. As before, with the fixed pair $w_1 = R_1R_2R_1R_0R_3R_4R_3R_2R_1R_4R_3R_4$ and $w_2 = R_0R_3R_4R_3R_2R_1R_4R_3R_4R_3$ of length twelve. Each of the fifteen nonzero entries of the difference of the linear parts carries the factor

$$\bigl(a_1a_3^2 - a_1a_4^2 - 2a_2a_3a_4\bigr)\bigl(a_1a_3^2 - a_1a_4^2 + 2a_2a_3a_4\bigr),$$

the remaining factors being monomials, nonzero on the positive orthant. The two branches give the displayed condition. This family is the mirror of Theorem 5.7 at the opposite end of the leg tuple, as the reversal symmetry of the path simplex requires. ☐

Remark 5.9 (the failure locus is a union of hypersurfaces)*.* Theorems 5.5, 5.7 and 5.8 account for *every* failure in an exhaustive scan of the 5748 primitive pairwise-distinct tuples with entries at most 12. There are 96 failures, 60 at depth 8 and 36 at depth 12, and none is left over. The same holds on the smaller box of entries at most 10, where there are 50 failures. Since Theorem 5.8 is Theorem 5.7 read at the reversed leg tuple, the description involves two loci, the second contributing one component per end of the path. The first two criteria were additionally frozen and tested out of sample on ten tuples outside the scanned box, with ten of ten deviating at the predicted depth, and the first against four controls off its locus, none of which deviates.

So in dimension four the tuples at which ρ_a is known to fail are not sporadic: they lie on explicit hypersurfaces in the leg space, cut out by $a_1a_3 = a_2a_4$ and by $a_4|a_1^2 - a_2^2| = 2a_1a_2a_3$, and on each the witness is a single word independent of the tuple. Whether the failure locus is exhausted by finitely many such hypersurfaces is open. This is the form in which we would expect an answer to Problem 11.13 in dimension four: not a single faithful tuple or a single counterexample, but a description of the locus.

Remark 5.10 (the loci are local to a window, and propagate)*.* The condition of Theorem 5.5 is not special to dimension four. It involves four consecutive legs, and in higher dimensions it can be read on any consecutive window of four. An exhaustive scan of the 1260 primitive pairwise-distinct tuples in $\mathbb{Q}^5_{>0}$ with entries at most 7 finds 48 failures, and every one has a vanishing window: 34

satisfy $a_1a_3 = a_2a_4$, on the first window, and 14 satisfy $a_2a_4 = a_3a_5$, on the second. None satisfies neither, and none satisfies both.

So the dimension-four families are local conditions on four consecutive legs, and the failure locus in dimension n contains the union of their translates along the leg tuple. This is consistent with W_3 being a standard parabolic of W_n on every four consecutive indices: a collision confined to such a window is inherited by every larger dimension.

Remark 5.11 (aligned valuations do not suffice). For an amalgam argument in dimension four one wants a tuple at which the splitting $W_4 = \langle R_0, R_2, R_4\rangle *_{\langle R_0,R_4\rangle} \langle R_0, R_1, R_3, R_4\rangle$, which exists because $\{0, 4\}$ is a separating clique of Γ_4, is matched by a p-adic valuation, so that one factor is integral and only the extra generator has a pole. At $(1, 2, 3, 4)$ this holds exactly, with $|m_1|^2 = 5$, $|m_2|^2 = 13$, $|m_3|^2 = 25$: at $p = 13$ every generator but R_2 is integral, and $v_{13}(R_2) = -1$. Nevertheless the point group at that tuple collapses at depth 12, and ρ_a is not injective there by Theorem 4.2. So aligning a valuation with the splitting is not sufficient for faithfulness, and an amalgam argument in dimension four needs an input that this alignment does not provide.

Remark 5.12 (the picture in dimension three). Theorems 4.2 and 5.2 are complementary and together settle the three-dimensional case. Faithfulness holds generically, and fails on a nonempty set of rational Class C tuples. So Class C is not the right condition: it neither implies faithfulness, by Theorem 4.2, nor is needed for it. The exceptional set is where the point group collapses, and by Corollary 4.4 that is exactly where ρ_a can fail.

Nothing here decides $n \geq 4$. The argument used that W_3 is virtually free, and that already fails at $n = 4$, though not for the reason one might expect. A triangle in Γ_n is no obstruction: it contributes a *finite* parabolic $(\mathbb{Z}/2)^3$, compatible with virtual freeness. The obstruction is an induced four-cycle. In Γ_4 the set $\{0, 3, 1, 4\}$ induces the cycle $0-3-1-4-0$, its non-edges being $\{0, 1\}$ and $\{3, 4\}$; a four-cycle is the join of two non-adjacent pairs, so the corresponding parabolic is $D_\infty \times D_\infty$, which contains $\mathbb{Z}^2$. Hence W_4 is not virtually free and Lemma 5.1 has no analogue there.

Remark 5.13 (a non-orthoscheme configuration for W_4 in $O(4)$). Because Problem 11.15 asks only for *some* embedding, the generators need not be reflections in the facets of an orthoscheme, and need not be reflections at all. The following configuration is not a point group of any orthoscheme and has a much simpler shape.

Split $\mathbb{R}^4 = P_1 \oplus P_2$ with $P_1 = \langle e_1, e_2\rangle$, $P_2 = \langle e_3, e_4\rangle$, and set

$$R_0 = \mathrm{diag}(-1, 1, 1, 1), \qquad R_4 = \mathrm{diag}(1, 1, -1, 1),$$

let R_1 be a reflection with normal in P_1, let R_3 be a reflection with normal in P_2, and let R_2 be a reflection whose normal lies in the *mixed* plane $\langle e_2, e_4\rangle$. Commuting with the two diagonal involutions forces any candidate for R_2 to preserve $\langle e_1\rangle$, $\langle e_3\rangle$ and $\langle e_2, e_4\rangle$; taking its normal inside the mixed plane is what breaks commutation with R_1 and with R_3, since those act inside P_1 and P_2 separately.

With normals along the Pythagorean directions $(3, 4)$, $(5, 12)$ and $(8, 15)$ the five involutions satisfy the defining relations of W_4 exactly, commuting precisely when $|i - j| \geq 2$, and the orbit growth agrees with the envelope through depth 12, computed in exact rational arithmetic with no modular reduction. The structure is transparent: R_0 and R_4 are diagonal, R_1 acts inside P_1, R_3 inside P_2, and R_2 is the only generator coupling the two planes.

This is not a proof that W_4 embeds in $O(4)$; agreement to a finite depth never is. What makes the configuration worth recording is its arithmetic, which separates the amalgam $W_4 = A *_C B$ with $A = \langle R_0, R_2, R_4\rangle \cong (\mathbb{Z}/2)^3$, $B = \langle R_0, R_1, R_3, R_4\rangle \cong D_\infty \times D_\infty$ and $C = \langle R_0, R_4\rangle \cong (\mathbb{Z}/2)^2$ across three distinct places.

The three normals carry the primes 5, 13 and 17 separately. Writing v_p for the minimum p-adic valuation of the entries, R_0 and R_4 are integral at all three, while R_1, R_3 and R_2 have $v_5 = -2$, $v_{13} = -2$ and $v_{17} = -2$ respectively and are integral at the other two. Consequently C is integral everywhere; all four elements of $A \setminus C$ have $v_{17} = -2$ with $v_5 = v_{13} = 0$; and the infinite part of B moves only the places 5 and 13, with $v_5\big((R_0R_1)^k\big) = -2k$ and $v_{13}\big((R_3R_4)^k\big) = -2k$ growing linearly while their valuations at the other two places remain exactly 0. The alternation $v_{17}\big((R_0R_2)^k\big) = -2, 0, -2, 0, \ldots$ is the finiteness of A, since R_0R_2 has order two.

So A and B move disjoint sets of places and C moves none, which is the separation an amalgam ping-pong requires. This is qualitatively stronger than the single-prime alignment of Remark 5.11, which held at the orthoscheme tuple $(1, 2, 3, 4)$ and did not prevent a collapse: there one prime had to separate both factors, whereas here each factor has its own. The ping-pong condition itself can be stated and tested. Writing an alternating word as $a_1b_1a_2 \cdots a_k$ with $a_i \in A\setminus C$ and $b_i \in B\setminus C$, the identity has $v_{17} = 0$, so it suffices that $v_{17} < 0$ for every such word. Over 1800 random alternating words with $1 \le k \le 9$ this holds in every case, and the value closest to zero drifts steadily away from it, the maxima over 200 trials at each k being $-2, -3, -4, -5, -7, -8, -10, -11, -14$.

The natural sharper guess is false and is recorded so that it is not attempted: v_{17} is *not* equal to $-2k$. Of 420 words tested, 176 deviate, because the elements of $B \setminus C$, while 17-integral, can cancel part of the 17-adic denominator built up by the a_i. So any proof must use the weaker monotonicity rather than exact additivity of valuations. We have not carried out the ping-pong.

There is a structural reason for this, and it rules out the whole family of arguments that would recover exact additivity. Suppose the normal of R_2 is $v = (0, e, 0, f)$ and p is chosen so that $p^2 \mid |v|^2$; this is what makes $R_2 = M/p^2$ with $M \equiv -2vv^T \pmod p$ of rank one, and rank one is what makes the numerator of an alternating word telescope into $(-2)^k(b_0v)(v^Tb_k)\prod_i(v^Tb_iv)$. Exact additivity would follow from

$$v^Tbv \not\equiv 0 \pmod p \qquad \text{for all } b \in B \setminus C. \tag{2}$$

But B preserves the splitting $\langle e_1, e_2\rangle \oplus \langle e_3, e_4\rangle$, so $v^Tbv = e^2b_{22} + f^2b_{44}$, and (2) evaluated at $b_{22} = b_{44} = 1$ reads $|v|^2 \not\equiv 0 \pmod p$, which is the negation of the condition that produced the pole. The identity itself lies in C and is excluded, but the image of each factor of $B \cong D_\infty \times D_\infty$ in $\mathrm{GL}_2(\mathbb{F}_p)$ is finite, the value 1 is attained by many elements of each, and the product is direct, so every pair of values occurs and (2) fails for infinitely many $b \in B \setminus C$. At $v = (0, 9, 0, 40)$ with $p = 41$ the two factors have order 10 and 40, with $b_{22} \in \{1, 3, 17\}$ and $b_{44} \in \{0, 1, 3, 9, 17, 19, 22, 24, 32, 38, 40\}$; the first set is contained in the second. Sampling does not detect this: a failure needs a coincidence in both blocks at once, and 141 sampled elements of $B \setminus C$ produced none. The obstruction is an identity in e, f, p, not a feature of the configuration, so no choice of normal avoids it. A proof must therefore either break the direct product structure of B, by placing R_1 and R_3 in planes that are not complementary, or use a seed whose degeneracy is not that of an isotropic normal.

6 The compact embedding theorem

The main new result of this paper is that a right-angled Coxeter group on N generators embeds in the *compact* group $O(N-1)$, one dimension below its Tits representation, whenever an explicit and widely satisfied arithmetic condition holds. The image is then dense rather than discrete, which is the opposite regime from the known constructions discussed in Remark 6.11. The mechanism is general: it applies to any irreducible, infinite, non-affine Coxeter system admitting a symmetric Cartan matrix of rank $|S| - 1$, and the right-angled case is where the remaining hypotheses can be verified.

We first prove the general dimension drop, then transfer it to a definite form, then specialise. The Lorentzian construction that first produced the four-dimensional case is retained in Section 6.5 as a special case, together with the proof that it stops there.

6.1 Vinberg's theorem without a signature hypothesis

The input is Vinberg's theorem in its general form. Call a symmetric matrix $A = (a_{ij})_{i,j=0}^n$ a *Cartan matrix* for (W_n, S) if $a_{ii} = 2$, if $a_{ij} = 0$ exactly when $|i-j| \geq 2$, and if $a_{ij}a_{ji} \geq 4$ whenever $|i-j| = 1$. Vinberg's theorem [1] states that the associated representation $\rho_A \colon W_n \to GL(V)$, $V = \mathbb{R}^{n+1}$, with $\rho_A(r_i)x = x - a_i^*(x)e_i$ and $a_i^*(e_j) = a_{ij}$, is faithful and has discrete image. No hypothesis is placed on the signature of A. With $A = 2G(c)$ the conditions read $c_i \geq 1$, which is exactly the Vinberg-admissible region of Section 6.6; what Theorem 6.21 forbids is only that the signature be Lorentzian.

Lemma 6.1 (the radical is a trivial line)**.** *Let $c \in [1,\infty)^n$ with $\det G(c) = 0$, and let B be the form of $G(c)$ on V. Then $V_0 = \operatorname{rad} B$ has dimension exactly 1, W_n acts trivially on V_0, and the induced form on V/V_0 is nondegenerate, with W_n acting by orthogonal reflections in the images $\bar{e}_0, \dots, \bar{e}_n$, whose Gram matrix is $G(c)$.*

Proof. $G(c)$ is a Jacobi matrix: tridiagonal with off-diagonal entries $-c_i \neq 0$. Jacobi matrices have simple spectrum, so $\dim\ker G(c) \leq 1$, and $\det G(c) = 0$ forces equality. For $x \in V_0$ we have $B(x, e_i) = 0$, hence $\rho(r_i)x = x - 2B(x, e_i)e_i = x$, so the action on V_0 is trivial and V_0 is W_n-invariant. The induced form is nondegenerate by definition of the radical, and $\rho(r_i)$ descends to $\bar{x} \mapsto \bar{x} - 2\bar{B}(\bar{e}_i, \bar{x})\bar{e}_i$ with $\bar{B}(\bar{e}_i, \bar{e}_i) = 1$, an orthogonal reflection. □

Lemma 6.2 (the quotient stays faithful)**.** *For $n \geq 2$ and c as in Lemma 6.1, the representation of W_n on V/V_0 is faithful.*

Proof. Let N be its kernel and fix a generator z of V_0. For $w \in N$ we have $\rho(w) = I + z\varphi_w^\top$ for some $\varphi_w \in V^*$, and $\rho(w)z = z$ by Lemma 6.1, so $\varphi_w(z) = 0$. Hence $\rho(w_1)\rho(w_2) = I + z(\varphi_{w_1} + \varphi_{w_2})^\top$, so $w \mapsto \varphi_w$ is a homomorphism and N is abelian; and $(I + z\varphi^\top)^k = I + kz\varphi^\top$, so every non-identity element of N has infinite order and N is torsion free. Since ρ is faithful on V by Vinberg's theorem, N is isomorphic to its image, so N is a torsion-free abelian normal subgroup of W_n.

The Coxeter diagram of W_n is the path on $n+1 \geq 3$ vertices with every edge labelled ∞. It is connected, so (W_n, S) is irreducible; it is infinite; and it is not affine, since no irreducible affine diagram on three or more vertices carries the label ∞. By Caprace–Fujiwara [2] such a Coxeter group acts on its Davis complex with a rank-one isometry and is therefore acylindrically hyperbolic, and by Osin [4] an acylindrically hyperbolic group has no infinite amenable normal subgroup. As N is torsion free, $N \neq 1$ would make it infinite and abelian, hence infinite amenable. So $N = 1$. □

6.2 The dimension drop and the compact form

Nothing in Lemma 6.2 refers to Γ_n, and Lemma 6.1 uses the path only to force the nullity to be one, which can be assumed instead. Separating the two gives the following, of which Theorem 6.13 is one instance.

Theorem 6.3 (dimension drop)**.** *Let (W, S) be an irreducible, infinite, non-affine Coxeter system with $|S| = N$, and let A be a symmetric Cartan matrix for (W, S) with $\operatorname{rank} A = N - 1$. Then W admits a faithful $(N-1)$-dimensional representation preserving a nondegenerate symmetric bilinear form; that is, W embeds in $O(p, q)$ with $p + q = N - 1$.*

Proof. Vinberg's theorem gives a faithful ρ_A on $V = \mathbb{R}^N$. The radical V_0 of the form has dimension $N - \operatorname{rank} A = 1$, and W acts trivially on it, as in Lemma 6.1. The proof of Lemma 6.2 applies verbatim, since it uses only that V_0 is a line with trivial action and that (W, S) is irreducible, infinite and non-affine. $\square$

Corollary 6.4. *If (W, S) is irreducible, infinite and non-affine and its Tits form has corank 1, then W embeds in $O(p, q)$ with $p + q = N - 1$.*

For right-angled systems the passage to the compact form also becomes general, and the existence of the compact point costs nothing.

Lemma 6.5 (compact points are automatic)**.** *Let Γ be a graph on N vertices whose complement is connected, and let C be the complement's adjacency matrix with arbitrary positive weights, scaled so that its largest eigenvalue is 1. Then $G = I - C$ is positive semidefinite of rank exactly $N - 1$.*

Proof. C has nonnegative entries and its underlying graph is connected, so by Perron–Frobenius its largest eigenvalue is simple. Scaling makes that eigenvalue 1, so $G = I - C$ is positive semidefinite with a one-dimensional kernel. $\square$

Theorem 6.6 (right-angled version)**.** *Let Γ be a graph on $N \geq 3$ vertices whose complement is connected, so that W_Γ is irreducible, infinite and non-affine. Suppose there are weights $c_e \geq 1$ on the complement edges with $\det(I - C(c)) = 0$, and that $\det(I - C(c))$ is irreducible, which by Theorem 6.9 holds whenever the complement has a bridge and at least two edges. Then W_Γ embeds in $O(N - 1)$.*

Proof. Let E be the edge set of the complement and $Y = \{c : \det(I - C(c)) = 0\}$, irreducible of dimension $|E| - 1$ by hypothesis. By Lemma 6.1 the condition that $\rho_c(w)$ act trivially on V/V_0 is $(\rho_c(w) - I)V \subseteq V_0(c)$, which is Zariski closed; write $Z_w \subseteq Y$ for that locus. Replacing c_e by $-c_e$ is conjugation by a diagonal sign change, so the condition depends only on c_e^2 and Z_w is well defined. At the assumed point with all $c_e \geq 1$, Theorem 6.3 gives faithfulness on the quotient, so that point lies in no Z_w with $w \neq 1$; hence each Z_w is a *proper* Zariski-closed subset of the irreducible Y.

By Lemma 6.5 the compact locus $K = \{c \in Y(\mathbb{R}) : I - C(c) \text{ is positive semidefinite of rank } N - 1\}$ is nonempty, and positive semidefiniteness with nullity one is an open condition, so K is open in $Y(\mathbb{R})$ of dimension $|E| - 1$ and therefore Zariski dense in Y. Consequently each $Z_w \cap K$ is closed with empty interior in K: otherwise it would be $(|E| - 1)$-dimensional, hence Zariski dense in the irreducible Y, forcing $Z_w = Y$. As W_Γ is countable and K is a Baire space, some $c \in K$ avoids every Z_w with $w \neq 1$. At that point $I - C(c)$ is the Gram matrix of N unit vectors in $\mathbb{R}^{N-1}$ with the prescribed orthogonality pattern; the reflections in them generate W_Γ inside $O(N - 1)$, faithfully. $\square$

Part of the irreducibility hypothesis can be discharged outright. Write $f_H(c) = \det(I - C(c))$ for a graph H with independent edge variables, so that f_H has constant term 1, has degree at most 2 in each c_e, and has $-f_{H-u-v}$ as the coefficient of c_e^2 for $e = uv$, by expansion of the determinant along the transposition (uv).

Lemma 6.7 (f_H is never a square)**.** *If H has at least one edge then f_H is not the square of a polynomial.*

Proof. Let $e = uv$ be an edge. The coefficient of c_e^2 in f_H is $-f_{H-u-v}$, whose constant term is -1; in particular $\deg_{c_e} f_H = 2$. If $f_H = s^2$ then $\deg_{c_e} s = 1$, say $s = \alpha c_e + \beta$ with α, β free of c_e, and comparing coefficients of c_e^2 gives $\alpha^2 = -f_{H-u-v}$. Setting every variable to zero yields $\alpha(0)^2 = -1$, impossible over $\mathbb{R}$. $\square$

Lemma 6.8 (coprimality). *If u is a non-isolated vertex of K, then* $\gcd(f_K, f_{K-u}) = 1$.

Proof. Induction on $|V(K)|$, the case of a single vertex being trivial since $f = 1$. Let $d = \gcd(f_K, f_{K-u})$. As d divides f_{K-u}, it lies in the polynomial ring on the edges of $K - u$ and so is free of the variables c_{uw}. Viewing f_K as a polynomial in those variables over that ring, d divides each coefficient. The coefficient free of them is f_{K-u}, and the coefficient of c_{uw}^2 is $-f_{K-u-w}$. Hence d divides $\gcd(f_{K-u}, f_{(K-u)-w})$ for any neighbour w of u, which is 1 by induction if w is non-isolated in $K - u$, and is 1 anyway if $f_{K-u} = 1$. $\square$

Theorem 6.9 (irreducibility for graphs with a bridge). *Let H be a graph with at least two edges having a bridge $e = uv$. Then f_H is irreducible over $\mathbb{Q}$.*

Proof. Since e lies on no cycle, no linear subgraph uses e other than as the transposition (uv), so $f_H = A - c_e^2 B$ with $A = f_{H-e}$ and $B = f_{H-u-v}$, both free of c_e. A factorisation of f_H splits its c_e-degree as $(2, 0)$ or $(1, 1)$. In the first case the factor free of c_e divides both A and B, and $\gcd(A, B) = 1$ as shown below, so that factor is constant. In the second, writing $f_H = (\alpha c_e + \beta)(\alpha' c_e + \beta')$ gives $\alpha\alpha' = -B$, $\beta\beta' = A$ and $\alpha\beta' + \alpha'\beta = 0$; multiplying the last by $\alpha'\beta$ yields $AB = (\alpha'\beta)^2$. Since $\gcd(A, B) = 1$ and $A(0) = B(0) = 1$, both A and B are then squares. But $H - e$ has an edge, because H has at least two, so $A = f_{H-e}$ is not a square by Lemma 6.7.

It remains to see $\gcd(A, B) = 1$. As e is a bridge, $H - e$ is the disjoint union of the component H_u of u and the component H_v of v, so $A = f_{H_u} f_{H_v}$ and $B = f_{H_u-u} f_{H_v-v}$, the two pairs involving disjoint sets of variables. An irreducible common divisor would divide one of f_{H_u}, f_{H_v}, say f_{H_u}, hence involve only variables of H_u, hence divide f_{H_u-u}; that contradicts Lemma 6.8. $\square$

This covers every graph whose complement has a bridge, in particular every tree, and so in particular the path P_{n+1} of Section 6.4. The graphs it does not cover are the 2-edge-connected ones, where every edge lies on a cycle and f_H acquires a term linear in c_e; there the factorisation criterion becomes a statement about a discriminant rather than about AB being a square, and we have not settled it. Among connected graphs on at most five vertices, 11 of the 21 isomorphism classes are 2-edge-connected, and all of them are irreducible by direct factorisation.

Remark 6.10 (the irreducibility hypothesis). For Γ_n the hypothesis holds, since the continuant is linear in x_n with coprime coefficients. In general it was checked by symbolic factorisation of $\det(I - C(c))$ over $\mathbb{Q}$, with the c_e independent indeterminates, for every isomorphism class of connected complement on at most six vertices: all 2, 6, 21 and 112 classes at $N = 3, 4, 5, 6$ give an irreducible polynomial. At $N = 2$ it fails, $\det = 1 - c^2 = (1 - c)(1 + c)$, but there W is infinite dihedral and affine, so it is excluded already. We expect irreducibility for every connected complement on at least three vertices and do not prove it.

The hypothesis cannot simply be dropped by relocating the Vinberg point. At any c with $c_e \geq 1$ one has $C(c) \geq C(1)$ entrywise, so $\lambda_{\max}(C(c)) \geq \lambda_{\max}(C(1)) > 1$, the last inequality because the Tits form of an irreducible infinite non-affine system is indefinite. A Vinberg point therefore never lies on the Perron sheet $\lambda_{\max} = 1$, which is exactly the compact locus: the two live on different sheets of the real hypersurface, and only irreducibility over $\mathbb{C}$ connects them.

The hypothesis on c is the only other one, and it holds widely. Over all right-angled systems with connected complement it was checked by exact evaluation of $\det(I - C)$: at $N = 4$ it holds for 33 of 38, at $N = 5$ for 723 of 728, and at $N = 6$ for 25348 of 26704. The failures are systematic rather than sporadic. When the complement is a star, $\det(I - C) = 1 - \sum_e c_e^2 \leq 1 - (N - 1)$ is negative throughout the region, so no such c exists. At $N = 3$ all four systems fail, which is forced: $O(2)$ is virtually solvable and each of the four groups contains a free group of rank two, so none of them embeds, and the criterion correctly detects this.

Beyond Γ_n, the conclusion was verified by enumeration at $N = 5$ for the complement of the five-cycle at $c = (1,1,1,1,2)$, for the tree with complement edges $01, 12, 13, 34$ at $c = (2,3,6,2)$, and for the path with a chord at the Tits point; in each case the quotient representation reproduces the growth series of W_Γ through depth 8, namely $1, 5, 15, 40, 105, 275, 720, 1885, 4935$ in the first and third cases and $1, 5, 14, 34, 82, 198, 478, 1154, 2786$ in the second.

6.3 Relation to known results

Remark 6.11 (relation to known results). Faithful representations of right-angled Coxeter groups into indefinite orthogonal groups are known and are not what is claimed here. The closest work is Danciger, Guéritaud and Kassel [3], whose Theorem 1.5 gives, for every irreducible right-angled Coxeter group on k generators, a smooth path of faithful discrete representations into $O(p, q+1)$ with $p + q + 1 = k$. Their construction in [3, §7.1] uses exactly the Cartan family of Section 6.2, specialised to the uniform line: they set $M_t(i, j)$ equal to 1, 0 or $-t$ according as m_{ij} is 1, 2 or ∞, with $t > 1$. They then observe that $\det M_t$ is a nonzero polynomial in t and restrict t to an interval avoiding its finite zero set, so that the form is nondegenerate of signature $(p, q+1)$ and the representation has dimension k.

Theorem 6.3 concerns exactly the locus they remove. A degenerate form is of no use for their purpose, which is proper discontinuity on the pseudo-Riemannian space $H^{p,q}$; here degeneracy is the hypothesis, and it is what produces a representation of dimension $k - 1$ rather than k. The families also differ in size: theirs is the one-parameter uniform line, ours the full weight space $(c_e)_e$ cut by $\det = 0$. The difference is not vacuous. Among connected complements on six vertices, 104 of the 112 isomorphism classes admit an admissible point in the full weight space, while only 75 have $\det M_t = 0$ for some $t > 1$ on the uniform line; so 29 classes are reached here and not by any specialisation of their parameter.

Theorem 6.6 is further from [3], and not by accident. Their program is proper discontinuity and discrete faithful images, and an infinite group has no infinite discrete image in a compact Lie group and no properly discontinuous action on one. Their signature is $(p, q+1)$ with $p \geq 1$ and $q + 1 \geq 1$ throughout, hence never definite; and their one dimension reduction, [3, Prop. 5.1], assumes Γ is already a cocompact reflection group of H^p, lands in the Lorentzian $O(p + c, 1)$, and reduces by a chromatic number rather than to $k - 1$. The definite case, where the image is dense rather than discrete, lies outside that framework.

Finally, the configuration space used in Theorem 6.6, unit vectors in $\mathbb{R}^N$ with a prescribed orthogonality pattern, is the object studied under the names orthogonal representation of a graph and minimum semidefinite rank, and Lemma 6.5 is a statement in that language.

6.4 Right-angled instances

Lemma 6.12 (admissible points exist)**.** *For every $n \geq 3$ there is $c \in [1,\infty)^n$ with $\det G(c) = 0$. For $n = 2$ there is none.*

Proof. Write $x_i = c_i^2 \geq 1$ and $D_0 = D_1 = 1$, $D_{k+1} = D_k - x_k D_{k-1}$, so $\det G(c) = D_{n+1}$. For $n = 2$, $D_3 = 1 - x_1 - x_2 \leq -1 < 0$. In general, if $D_{n-1} \neq 0$ then $x_n = D_n/D_{n-1}$ solves $D_{n+1} = 0$, and this is admissible precisely when $D_n/D_{n-1} \geq 1$. Taking $x_{n-1} = 1$ gives $D_n = D_{n-1} - D_{n-2}$, so the requirement is $-D_{n-2}/D_{n-1} \geq 0$, that is, D_{n-2} and D_{n-1} of opposite signs, or $D_{n-2} = 0$.

Prescribe the first entries and let the rest be 1; the recursion $D_{k+1} = D_k - D_{k-1}$ then has period 6. For $x = (1, 1, \dots)$ the values of D_k are $1, 1, 0, -1, -1, 0$ repeating, so $D_{n-2} = 0$ exactly when $n \equiv 1 \pmod 3$. For $x = (2, 1, 1, \dots)$ they are $1, -1, -2, -1, 1, 2$ repeating from $k = 1$, and consecutive values differ in sign exactly at $k \equiv 1, 4 \pmod 6$, giving $n \equiv 0 \pmod 3$. For

$x = (2, 1, 3, 1, 1, \dots)$ they are $1, 3, 2, -1, -3, -2$ repeating from $k = 4$, with sign changes at $k \equiv 0, 3 \pmod 6$, giving $n \equiv 2 \pmod 3$. The three cases cover all $n \geq 3$. $\square$

The resulting points were checked in exact rational arithmetic for every n from 3 to 200: $\det G = 0$ in every case, with x_n equal to 1, 2 and 3 in the three families respectively.

Lemma 6.2 was also tested directly. Rescaling the Cartan matrix to $a_{i,i+1} = -2x_i$ and $a_{i+1,i} = -2$ is conjugation by a diagonal matrix, so it presents the same representation over $\mathbb{Q}$, and the quotient may be enumerated exactly. Its sphere sizes are compared with the right-angled Coxeter growth series $1/f_\Gamma(-t/(1+t))$, where f_Γ is the clique polynomial of Γ_n:

$$\begin{array}{lll} n = 4, & f = 1 + 5x + 6x^2 + x^3, & 1, 5, 14, 33, 75, 169, 380, 854, 1919, 4312, 9689, 21771, \\ n = 5, & f = 1 + 6x + 10x^2 + 4x^3, & 1, 6, 20, 54, 136, 334, 812, 1966, 4752, 11478, 27716, \\ n = 6, & f = 1 + 7x + 15x^2 + 10x^3 + x^4, & 1, 7, 27, 82, 226, 597, 1545, 3957, 10080, 25605. \end{array}$$

In each case the enumerated quotient reproduces the series exactly to the depth shown, the $n = 5$ and $n = 6$ rows being the informative ones since their signatures $(3, 2, 1)$ and $(4, 2, 1)$ admit no hyperbolic geometry. A single collapsed word would have shortened some sphere.

Theorem 6.13 (the embedding, in every dimension)**.** *For every $n \geq 3$ the group W_n embeds in $O(n)$, and hence in $\mathrm{Isom}(\mathbb{R}^n)$. For $n = 2$ it does not.*

Proof. Apply Theorem 6.6 to Γ_n, whose complement is the path P_{n+1}, connected for every $n \geq 1$. Lemma 6.12 supplies a point with all $c_i \geq 1$ and $\det G(c) = 0$. The irreducibility hypothesis holds here: writing $x_i = c_i^2$, the continuant D_{n+1} is linear in x_n with coefficients D_n and $-D_{n-1}$, and consecutive continuants are coprime, since a common factor of D_n and D_{n-1} divides $x_{n-1}D_{n-2}$ and one descends; so D_{n+1} is irreducible. Composing $O(n) \hookrightarrow \mathrm{Isom}(\mathbb{R}^n)$ gives the embedding.

For $n = 2$, suppose $W_2 \hookrightarrow O(2)$. The subgroup of even-length words maps into $SO(2)$, which is abelian. But with $a = r_0 r_1$ and $b = r_1 r_2$ the even subgroup is generated by a and b subject only to $(ab)^2 = 1$, since $r_0 r_2 = ab$ and the sole defining relation of W_2 is $(r_0 r_2)^2 = 1$; setting $t = ab$ presents it as $\langle a, t \mid t^2 = 1 \rangle \cong \mathbb{Z} * \mathbb{Z}/2$, which is not abelian. So no such embedding exists, and by Proposition 3.2 none into $\mathrm{Isom}(\mathbb{R}^2)$ either. $\square$

Remark 6.14 (why $n = 2$ is the exception)*.* The two halves of the proof fail together, which is the check that the argument is not lossy. Lemma 6.12 produces no admissible point at $n = 2$ because $D_3 = 1 - x_1 - x_2$ cannot vanish for $x_i \geq 1$, and independently W_2 genuinely does not embed. The same dichotomy explains $n = 1$: there $W_1 = D_\infty$ is affine, Lemma 6.2 does not apply, and indeed D_∞ embeds in $\mathrm{Isom}(\mathbb{R})$ only by using translations, its linear part having kernel $\mathbb{Z}$.

6.5 The Lorentzian route in dimension four

The valuation route above is closed, but the question it was aimed at is not. We settle it by a different mechanism, which avoids ping-pong entirely: produce one faithful four-dimensional representation over an *indefinite* form, where discreteness is classical, and then transport faithfulness to the compact form through the complex variety that contains both.

The mechanism rests on an accident of the diagram, and it is worth isolating first. The complement of Γ_n is the path on $n + 1$ vertices, so the Tits form of W_n is $G_n = I - A(P_{n+1})$, with eigenvalues $1 - 2\cos\big(k\pi/(n + 2)\big)$ for $1 \leq k \leq n + 1$.

Lemma 6.15 (the Lorentzian dimension)**.** *G_n has a one-dimensional radical if and only if $n \equiv 1 \pmod 3$, and exactly one negative eigenvalue if and only if $n \leq 4$. Both hold simultaneously only for $n = 4$, where the signature is $(3, 1, 1)$.*

Proof. A zero eigenvalue means $2\cos\bigl(k\pi/(n+2)\bigr) = 1$, that is $k\pi/(n+2) = \pi/3$, so $3k = n+2$ for exactly one k in range, which happens precisely when $n \equiv 1 \pmod 3$. A negative eigenvalue means $2\cos\bigl(k\pi/(n+2)\bigr) > 1$, that is $k < (n+2)/3$, and the number of such k is one exactly when $(n+2)/3 \le 2$. □

So for $n = 4$, and for no other n, the quotient of the Tits representation by its radical is n-dimensional and Lorentzian. That quotient is the representation we deform. It is convenient to work not at the Tits point itself but in a one-parameter family through it.

Lemma 6.16 (the admissible family)**.** *Let $G(t)$ be the symmetric 5×5 matrix with unit diagonal whose only nonzero off-diagonal entries are $G_{01} = G_{34} = G_{23} = -1$ and $G_{12} = -t$. Then for every real t,*

$$\det\bigl(G(t) - xI\bigr) = a(a-1)(a+1)\bigl(a^2 - (2+t^2)\bigr), \qquad a = 1 - x,$$

so the eigenvalues of $G(t)$ are 0, 1, 2, $1 \pm \sqrt{2+t^2}$, the signature is $(3,1,1)$, and the radical is spanned by $(1,1,0,-t,-t)$.

Proof. Expanding the tridiagonal determinant by the continuant recursion with diagonal a and off-diagonals $-1, -t, -1, -1$ gives $D_1 = a$, $D_2 = a^2 - 1$, $D_3 = a^3 - (1+t^2)a$, $D_4 = a^4 - (2+t^2)a^2 + 1$ and $D_5 = aD_4 - D_3 = a\bigl(a^4 - (3+t^2)a^2 + (2+t^2)\bigr)$, and the quartic factors as $(a^2-1)\bigl(a^2 - (2+t^2)\bigr)$. Since $2+t^2 > 1$ the root $1 - \sqrt{2+t^2}$ is negative and $1 + \sqrt{2+t^2}$ is positive, giving three positive eigenvalues, one negative and one zero. The stated vector is annihilated by $G(t)$ by direct substitution. □

For $t \ge 1$ every off-diagonal entry of $G(t)$ is either 0 or at most -1. In Vinberg's criterion an entry 0 is a right angle and an entry ≤ -1 is a pair of disjoint or tangent walls, both carrying $m_{ij} = \infty$; the resulting labels are exactly $m_{ij} = 2$ for $|i-j| \ge 2$ and $m_{ij} = \infty$ otherwise, which is the Coxeter presentation of W_4. By Lemma 6.16 the form is Lorentzian of rank four, so Vinberg's theorem [1] applies and the group generated by the five reflections is discrete with the associated polyhedron as fundamental domain, the natural map from W_4 being an isomorphism onto it.

Lemma 6.17 (faithful Lorentzian point)**.** *For each $t \ge 1$ the representation $\rho_t \colon W_4 \to O(3,1)$ determined by $G(t)$ is faithful. For $t > 1$ it moreover satisfies*

$$\det \mathrm{Gram}(v_1, v_2) = 1 - t^2 \neq 0, \qquad \det \mathrm{Gram}(v_2, v_3, v_4) = -1 \neq 0.$$

Proof. Faithfulness is the previous paragraph. The first determinant is immediate. For the second, the Gram matrix of (v_2, v_3, v_4) has unit diagonal, entries -1 in the positions $(2,3)$ and $(3,4)$, and 0 in the position $(2,4)$ because $|2-4| \ge 2$; its determinant is -1 independently of t. □

The non-degeneracy in Lemma 6.17 is what the Tits point itself fails: at $t = 1$ the pair v_1, v_2 spans a degenerate plane, which is why the family is introduced.

Theorem 6.18 (W_4 embeds in $O(4)$)**.** *There exist five reflections in $O(4)$ generating a group isomorphic to W_4. Consequently W_4 embeds in $\mathrm{Isom}(\mathbb{R}^4)$, and Problem 11.15 has an affirmative answer for $n = 4$.*

Proof. Let X be the affine variety of five-tuples of reflections of $\mathbb{C}^4$, equivalently of unit normals up to sign, whose normals satisfy $\langle v_i, v_j \rangle = 0$ for the six pairs with $|i-j| \ge 2$. Order the construction as v_1, v_2, v_4, v_3, v_0. Choosing v_1 and v_2 freely on the quadric $\langle v, v \rangle = 1$ contributes $3+3$; then $v_4 \perp v_1, v_2$ lies on the conic in the plane $\langle v_1, v_2 \rangle^{\perp}$, contributing 1; then $v_3 \perp v_1$ lies on the quadric in a three-space, contributing 2; and finally $v_0 \perp v_2, v_3, v_4$ is determined up to sign, contributing

0. Let X° be the locus where $\mathrm{Gram}(v_1, v_2)$ and $\mathrm{Gram}(v_2, v_3, v_4)$ are non-degenerate, so that each step has the stated dimension. Every fibre in the chain is a smooth affine quadric of dimension at least one modulo ± 1, hence irreducible, so X° is irreducible of dimension 9, in agreement with the naive count $5 \cdot 3 - 6$.

By Lemma 6.17 the Lorentzian configuration at any $t > 1$, read in a basis in which the complex form is standard, is a point P_t of X° at which the representation is faithful. Hence for each $w \in W_4 \setminus \{1\}$ the set $X_w = \{x \in X^\circ : \rho_x(w) = 1\}$ is Zariski-closed and proper.

Let X_c be the set of real points of X° for the positive definite form. It is non-empty: take $v_0 = e_1$, v_1 in $\langle e_1, e_2 \rangle$, v_3 and v_4 in $\langle e_3, e_4 \rangle$, and v_2 in $\langle e_2 \rangle \oplus \big(\langle e_3, e_4 \rangle \cap v_4^\perp\big)$, which satisfies the six orthogonality conditions by construction and has three free angles; together with the six dimensions of $O(4)$ this is a smooth real locus of dimension 9. Being a real form of full dimension inside the smooth irreducible X°, it is Zariski-dense in X°. Therefore each $X_w \cap X_c$ is a proper closed subset of X_c, in particular closed with empty interior. As W_4 is countable, $\bigcup_{w \neq 1}(X_w \cap X_c)$ is meagre, so by the Baire category theorem its complement in X_c is non-empty. Any point of that complement is a five-tuple of reflections of $\mathbb{R}^4$ generating a faithful copy of W_4. The last statement follows since $O(4) \subset \mathrm{Isom}(\mathbb{R}^4)$. ☐

Remark 6.19 (numerical corroboration). The proof gives no explicit faithful point, only that faithful points are generic. As a check on the bookkeeping, the compact configuration with normals along the Pythagorean directions of Section 6.5 was expanded over $\mathbb{F}_p$ with $p = 2^{31} - 1$. The six commuting relations hold and the four remaining pairs fail to commute, as required, and the sphere sizes

$$1,\ 5,\ 14,\ 33,\ 75,\ 169,\ 380,\ 854,\ 1919,\ 4312,\ 9689,\ 21771,\ 48919$$

agree with the growth series $1/f_{\Gamma_4}(-t/(1+t))$ of W_4 through depth 12, where $f_{\Gamma_4}(x) = 1 + 5x + 6x^2 + x^3$ is the clique polynomial. The last three values were predicted from the series before the corresponding depths were computed.

6.6 The route stops at four

Lemma 6.15 concerns only the Tits point, where all the walls are tangent. The proof of Theorem 6.18 did not use that point, but a deformation of it, so ruling out $n \geq 5$ requires ruling out the whole admissible region rather than one point. That is what we do here. The result is negative and complete: the mechanism of Theorem 6.18 is available in dimension four and in no other dimension.

Since the six relations $\langle v_i, v_j \rangle = 0$ for $|i - j| \geq 2$ say exactly that the Gram matrix is tridiagonal, every candidate is $G(c) = I - C(c)$, where $C(c)$ is the adjacency matrix of the path P_{n+1} with weights $c_1, \ldots, c_n$ on its edges $e_i = \{i-1, i\}$. Vinberg's criterion requires $c_i \geq 1$ for every i, since the label is $m_{ij} = \infty$ on each path edge, and the argument requires signature $(n-1, 1, 1)$: one negative eigenvalue to make the form Lorentzian, and one zero to make the rank n rather than $n+1$.

Lemma 6.20 (isolation)**.** *Suppose $G(c)$ has signature $(n-1,1,1)$ with all $c_i \geq 1$. If $c_i > 1$, then P_{n+1} contains no three consecutive vertices disjoint from and non-adjacent to e_i. Consequently $n - 3 \leq i \leq 4$.*

Proof. Suppose such a triple $\{a-1, a, a+1\}$ exists. The principal submatrix of $G(c)$ on the five vertices $\{i-1, i\} \cup \{a-1, a, a+1\}$ is block diagonal, because no edge joins the two groups. The first block has eigenvalues $1 \pm c_i$, so $1 - c_i < 0$. The second block is $I - C(P_3)$ with weights $u, v \geq 1$, whose eigenvalues are 1 and $1 \pm \sqrt{u^2 + v^2}$, and $\sqrt{u^2+v^2} \geq \sqrt{2} > 1$, so it contributes a second strictly negative eigenvalue. By Cauchy interlacing the second smallest eigenvalue of $G(c)$ is at most the second smallest of the submatrix, hence strictly negative. But signature $(n-1,1,1)$

forces the second smallest eigenvalue of $G(c)$ to be 0, a contradiction. For the last statement, a triple to the left of e_i exists as soon as $i \geq 5$, and one to the right as soon as $i \leq n-4$; failing both means $i \leq 4$ and $i \geq n-3$. $\square$

Theorem 6.21 (no admissible configuration beyond four)**.** *For every $n \geq 5$ there is no $c \in [1,\infty)^n$ with $G(c)$ of signature $(n-1,1,1)$.*

Proof. Write $D_0 = D_1 = 1$ and $D_{k+1} = D_k - c_k^2 D_{k-1}$, so that D_k is the k-th leading principal minor of $G(c)$ and $\det G(c) = D_{n+1}$. Signature $(n-1,1,1)$ requires $\det G(c) = 0$.

By Lemma 6.20 every index i with $c_i > 1$ satisfies $n-3 \leq i \leq 4$. For $n \geq 8$ this range is empty, so $c = (1,\ldots,1)$ is the Tits point, whose eigenvalues are $1 - 2\cos\bigl(k\pi/(n+2)\bigr)$; the number of negative ones is $\#\{k : k < (n+2)/3\} = \lceil (n-1)/3 \rceil \geq 3$, so the signature is wrong.

For $n = 5, 6, 7$ the range leaves at most the indices $\{2,3,4\}$, $\{3,4\}$ and $\{4\}$, so the continuant may be evaluated in closed form:

$$\begin{array}{llll} n=5, & c=(1,t,u,v,1), & \det G = t^2v^2, & \text{never } 0 \text{ for } t,v \geq 1,\\ n=6, & c=(1,1,t,u,1,1), & \det G = 1, & \text{never } 0,\\ n=7, & c=(1,1,1,t,1,1,1), & \det G = 1-t^2, & \text{zero only at } t=1. \end{array}$$

In the first two cases the rank never drops and the required zero eigenvalue is absent. In the third the only admissible parameter is $t = 1$, again the Tits point, whose signature is $(5,2,1)$ by Lemma 6.15. No case survives. $\square$

Remark 6.22 (what Theorem 6.21 does not say)*.* Theorem 6.21 closes one route, not the question. It says that no admissible configuration is *Lorentzian* beyond $n = 4$, and the Lorentzian condition was used only to invoke Vinberg's criterion in the form that produces a discrete hyperbolic reflection group. That form is stronger than necessary. Vinberg's theorem on linear Coxeter groups requires no condition on the signature at all, and dropping the requirement settles Problem 11.15 completely.

Remark 6.23 (what replaced the Lorentzian hypothesis)*.* Theorem 6.18 obtained $n = 4$ from a hyperbolic reflection group, which needs signature $(n-1,1)$, and Theorem 6.21 shows no other n supplies one. The signature was doing one job: guaranteeing that the n-dimensional quotient of the $(n+1)$-dimensional Vinberg representation stays faithful. Lemma 6.2 does that job directly, by observing that the kernel of the quotient is forced to be torsion-free abelian and that W_n has no such normal subgroup. The signatures that actually occur are $(n-1-k,k,1)$ with $k = \lceil (n-1)/3 \rceil$, so for $n \geq 5$ the intermediate group is $O(p,q)$ with $q \geq 2$ and no hyperbolic geometry is available; the argument does not need any.

Remark 6.24 (what the reduction does not buy)*.* It does not decide Problem 11.15 for any $n \geq 4$, the case $n = 3$ being settled affirmatively by Corollary 5.3, and by itself it does not touch Problem 11.13: an abstract embedding of W_n into $O(n)$ need not be by reflections, and the faithfulness of the particular map ρ_a is a separate question. The two standard invariants remain undecisive. Finite subgroups do not separate: every finite subgroup of W_n is conjugate into an elementary abelian 2-group on a clique of Γ_n, of rank at most $\lceil (n+1)/2 \rceil \leq n$, while $O(n)$ contains $(\mathbb{Z}/2)^n$. Free subgroups do not separate either, both groups containing free subgroups of rank two for $n \geq 3$.

7 The planar case

Let $F = C_2 * C_2 * C_2$ and $\rho_\tau \colon F \to G_\tau$ the generic triangle group.

Theorem 7.1 (Census; PROVED, computational input VERIFIED)**.** *There are* 33 *pairs of reduced words of length* 11*, distinct in* F*, with* $\rho_\tau(w) = \rho_\tau(w')$ *for every* τ*. Outside a proper Zariski-closed subset of shape space these are the only coincidences in the ball of radius* 11*, and there are* 132 *further pairs at radius* 12*. The generic sphere sizes are*

$$1, 3, 6, 12, 24, 48, 96, 192, 384, 768, 1536, 3039, 6012, 11925, 23570.$$

The 33 identities are polynomial identities in $\mathbb{Z}[p, q]$; the lower bounds come from counts in a finite field chosen to contain the needed roots of unity and $\sqrt{-1}$, and match the upper bounds from the identities, so the counts are exact.

Theorem 7.2 (Right-triangle structure; PROVED)**.** *For* $W = \langle R_0, R_1, R_2 \mid R_i^2 = (R_0R_1)^2 = 1\rangle$ *the rotation subgroup is of index* 4 *and isomorphic to* $\mathbb{Z}\wr\mathbb{Z}$*, with translation lattice* $\mathcal{T} = 2(t-1)\mathbb{Z}[t^{\pm 1}]$ *obtained from Crowell's exact sequence and the Bass–Serre tree of the free product. The universal sequence is shape-independent through depth* 32*, and the shape* $(1, 2)$ *deviates at depth* 33*. If* $c_T = N(a - ib) > 2d$ *then no deviation occurs at depth* d*.*

Remark 7.3 (status of the metric). Word length equals relaxed length plus twice the connectivity defect, $\ell_T = \ell_R + 2c$. The upper bound is PROVED. The lower bound is open and is stated as a conjecture; the closed form c_{pred} computed by the explicit transducer agrees with the true defect with 0 exceptions on the 275,823 elements of the ball of radius 24, held out at depths $19, 21, 23, 24$, which is HEURISTIC support and not a proof. The junction site cost is $\max(|d_L - 1|, |d_R|)$, which depends on the sign of d_L and not on its magnitude alone; the form $\max(|d_L| - 1, |d_R|)$ is false on every cell with $d_L \leq -2$ and $|d_R| \leq |d_L|$. This is a term inside the site-cost chain of the strand model, not an additive correction to the length formula.

8 The growth rate β_2

With $q = x^2$ and Σ_1 the odd k-recursion sum, $1 - \Sigma_1(q)$ is a Hahn–Exton q-cosine, and $q^* = 0.449453630558948046\ldots$ is its smallest positive zero.

Theorem 8.1 (PROVED)**.** *The growth rate of the planar right-triangle group is* $\beta_2 = 1/\sqrt{q^*} = 1.4916177871\ldots$*, and* $\beta_2 < 3/2$*.*

The proof squeezes the true count between the pure-travel and relaxed counts, which share the exponential rate, and locates q^* by a contour argument with exact rational winding number. Irrationality of β_2 is open (Remark 1.4).

9 Transcendence: the unconditional part

Theorem 9.1 (PROVED)**.** Σ_1 *and* S_1 *are transcendental over* $\mathbb{Q}(q)$ *and have* $|q| = 1$ *as a natural boundary.*

Theorem 9.2 (Travel resolvent; PROVED)**.** $\Sigma_0/(1 - \Sigma_1)$ *is transcendental over* $\mathbb{Q}(x)$*, unconditionally.*

Transcendence of V is conditional on (M) and $(R-J)$; transcendence of U is conditional on (M), $(R-J)$ and (T). Neither is claimed here.

10 The q-cosine increment

Theorem 10.1 (PROVED)**.** *For every algebraic $q \in (0,1)$ the difference Galois group of $G(q,z) = {}_1\phi_1(0;q;q^2,z)$ over $\mathbb{C}(z)$ contains SL_2, so G is hypertranscendental. The connection constant is $C_1 = 1/(q;q)_\infty$, whence $\prod_{k\ge 1} z_k(q) q^{2(k-1)} = (q;q^2)_\infty$.*

The general hypertranscendence statement, for all q not a root of unity and over $\bigcup_j \mathbb{C}(z^{1/j})$, is due to Adamczewski, Dreyfus and Hardouin and is stronger. The increment is the removal of a transcendence hypothesis on q in the SL_2 statement.

11 The five directions

Problem 11.1 (Prove $(R-J)$)**.** Show $\Pi_y(q_m) \neq 0$ for infinitely many travel poles. This would make V, and with (T) also U, transcendental over $\mathbb{Q}(x)$ unconditionally, modulo (M). What exists is an upper bound $|P_{12}|w^3 < 2$; what is needed is a lower bound $|P_{12}(q_m)| \ge c\, w_m^{-3}$.

Remark 11.2 (the travel problem is a Jacobi eigenvalue problem in a spectral gap)*.* The trailing coefficient of the three-term recursion is the constant $-q$, so the recursion symmetrises. Writing $q = r^2$ and $A_s = r^s z_s$ turns $A_{s+2} = p_s A_{s+1} - qA_s$ into

$$z_{s+2} + z_s \;=\; d_s\, z_{s+1}, \qquad d_s = \frac{p_s}{r} = \left(r + \frac{1}{r}\right) - \frac{e_s}{r}, \qquad e_s = 2q^{2+2s}(1-q) > 0,$$

a Jacobi recursion with *unit* off-diagonals. The free Jacobi operator with unit off-diagonals has spectrum $[-2,2]$, and $r + 1/r - 2 = (r-1)^2/r > 0$ for $0 < r < 1$, so the constant part of the diagonal lies strictly above the edge of that spectrum while the correction e_s/r decays geometrically. The travel poles are therefore eigenvalues in the spectral gap.

That is the regime in which discrete Sturm oscillation theory counts nodes, and it explains the observation that the null vector at the m-th pole has exactly $m-1$ sign changes. What it does not supply is the amplitude comparison: the sign pattern of R follows from oscillation theory, whereas $(S1)$ asks in addition that the last segment overshoot, $|\text{tail}| > |\Sigma_0|$, which is quantitative. The substitution and the gap inequality are verified in `lean/with_mathlib/TransferChain.lean`.

Proposition 11.3 (the constants of $(S1)$, in the continuum)**.** *Let $A(u) = \Sigma_0 \cos(wu)$ be the continuum form of the travel block, $u = q^s$, and let the travel null vector be $R_s = A_{s+1} - A_s$. Split R at its last sign change. Then*

$$\text{tail} = 2\,\Sigma_0, \qquad \text{head} = -\,\Sigma_0.$$

In particular $|\text{tail}| = 2|\Sigma_0| > |\Sigma_0|$, which is the overshoot $(S1)$ requires, with a factor of two to spare.

Proof. To leading order $R_s = A_{s+1} - A_s = -\tau u A'(u) = \tau u w\, \Sigma_0 \sin(wu)$, and a sum over s becomes $\int \cdot\, du/(\tau u)$, in which the factor τu cancels. Hence for any subinterval $I \subseteq (0,1]$ the corresponding partial sum is $w\Sigma_0 \int_I \sin(wu)\, du$.

The zeros of $\sin(wu)$ in $(0,1]$ are at $u = k\pi/w$, and the last sign change of R, that is the one at largest s, is the one at smallest u, namely $u = \pi/w$. So

$$\text{tail} = w\Sigma_0 \int_0^{\pi/w} \sin(wu)\, du = w\Sigma_0 \cdot \frac{1-\cos\pi}{w} = 2\Sigma_0,$$

independently of w. For the total, $w\Sigma_0 \int_0^1 \sin(wu)\, du = \Sigma_0(1-\cos w)$, and $\cos w = 0$ because $A(1) = A_0 = 0$ is the initial condition of the recursion; so the total is Σ_0, consistent with $\sum_s R_s = \Sigma_0$, and head $= \Sigma_0 - 2\Sigma_0 = -\Sigma_0$. □

Remark 11.4 (what this changes in the dependency chain). The computation above uses the continuum form of the travel block and nothing else. In particular tail $= 2\Sigma_0$ is independent of w, so it does not use the quantisation at all; and the one place $\cos w = 0$ enters is the exact initial condition $A_0 = 0$, not the sharp size of T_2, which fixes *which* root w_m is rather than that $\cos w$ vanishes.

So $(S1)$ does not depend on the flat-saddle estimate. What it does depend on is a discrete-to-continuum error bound for the travel recursion, and by Remark 11.2 that recursion is a Jacobi recursion with unit off-diagonals whose spectral parameter sits in a gap, which is the standard setting for Liouville–Green estimates for difference equations. That is a different, and more standard, analytic problem than the one blocking the quantisation. The numbers agree: the derived 2 and -1 against measured $1.9999991, 2.0000123, \ldots$ and $0.99999908, 1.0000123, \ldots$ at the poles where the measurement is clean.

Remark 11.5 (where the chain terminates). The measured laws for the two sides of $(S2)$ reduce, on unification, to the single scale $w = \sqrt{2/\tau}$ at a travel pole, together with the quantisation $w_m = (m - \frac{1}{2})\pi$. That quantisation is not an independent statement. Writing $S_e = \cos W - T_2$, the size of T_2 is $(\sqrt{2}/36)\sqrt{\tau}\,\sin X + O(\tau)$ uniformly in the phase, so the zeros of S_e are those of $\cos W$ displaced by $O(\sqrt{\tau})$, which is exactly $w_m = (m - \frac{1}{2})\pi + O(\sqrt{\tau})$.

So the chain reads: $(R{-}J)$ follows from $(S1)$ and $(S2)$, and $(S2)$ is the inequality $|C_\lambda| > xB$, whose measured margin is $w/2$.

It is worth being precise about what this does *not* need. An earlier version of this remark asserted that the chain terminates at the sharp size of T_2, the statement that fixes $w_m = (m - \frac{1}{2})\pi$. That is incorrect. Proposition 11.3 derives the constants of $(S1)$ using only the continuum form and the exact initial condition $A_0 = 0$; the quantisation enters nowhere, and the identity tail $= 2\Sigma_0$ is independent of w altogether. And $(S2)$ is an inequality, so it needs a lower bound on the ratio together with $w \to \infty$ along the poles; the latter is the infinitude of the zeros of $1 - \Sigma_1$ in $(0, 1)$, which the companion paper proves from the crude bound $|T_2| \to 0$, not from the sharp one. The sharp estimate fixes *which* root the mth pole is, and the chain never uses that.

What both $(S1)$ and $(S2)$ do need is an amplitude bound: for $(S1)$ a discrete-to-continuum error estimate, for $(S2)$ a lower bound on $|C_\lambda|/(xB)$ growing in w. By Remark 11.2 the underlying recursion is a Jacobi recursion with unit off-diagonals whose spectral parameter sits in a gap, which is the standard setting for Liouville–Green estimates for difference equations.

Remark 11.6 (two observations, and a route that does not work). First, the obstruction is a rank count, not a sign. The junction matrix $q^{\max(|a_L|-1, |a_R|)}$ has rank 4 on the tested block, and 6 after sign symmetrisation, so Π_y is a sum of several products rather than the single product that a rank-one kernel would give, and the terms can in principle cancel. At the same time the positivity does not degrade as quickly as one might expect from the interlacing of the zeros of S_e with those of $1 - \Sigma_1$: at the first six travel poles the dressed bulk values B_V and B_U are positive and increasing, $3.71,\ 493,\ 1979,\ 1.17 \cdot 10^4,\ 2.31 \cdot 10^4,\ 6.67 \cdot 10^4$ for B_V, and the gate signs $b_0 > 0$, $\mathfrak{g}_V t_1 < 1$ hold at every one.

Second, the sign of the travel null vector is a red herring, and this appears to be missed in the source. Since $L_s = R_s/e_s$, both factors of $\Pi_y = \langle \lambda, R \rangle \langle L, \mu \rangle$ are linear in R, so Π_y is a *quadratic form* in R. The vectors λ and μ are built from the bulk and do not involve R, so the overall sign and normalisation of R cancel in the product. That sign is precisely what flips when q_m crosses a zero of S_e: the values $\Sigma_0(q_m) = \sum_s R_s$ alternate, being $1.156,\ -4.545,\ 7.752,\ -10.923,\ 14.080,\ -17.232,\ 20.381,\ -23.528,\ 26.6$ at the first nine poles, with $|\Sigma_0(q_m)|$ increasing by $3.389,\ 3.207,\ 3.171,\ 3.157,\ 3.152,\ 3.149,\ 3.147,\ 3.145$, so $\Sigma_0(q_m) \sim (-1)^{m+1}\pi m$. The alternation is real, and it does not reach Π_y. This observation is independent of how λ and μ are normalised, since it uses only that neither depends on R.

Numerically $\Pi_y(q_m) > 0$ at each of the first nine travel poles, the two factors sharing the sign of R throughout. The travel null vector is computed from the forward recursion $R_s = e_s(q^s A_s + B_s)$ with $A_0 = 0$, $B_0 = 1$, whose consistency condition $B_\infty = 0$ is exactly $\Sigma_1(q) = 1$; the bulk solve is the gap-marked bulk system, whose gap term $q^{a+b}\mathfrak{g}$ is rank one and so is handled by a Sherman–Morrison update. Both are validated against the published certificate: $\sum_s R_s$ reproduces $\Sigma_0(q_m)$ to ten digits, and B_V, B_U reproduce 3.7105027, 3.3241016, 492.95283, 233.43575, 1978.5952, 897.12122, 11663.55, 5110.5716 exactly. This is evidence, not proof: nine poles are not infinitely many, and the identification of the far-junction vector λ used here agrees with μ, which reproduces the proved positivity at the dominant pole but is not independently derived.

Third, the mechanism behind that positivity is a Sturm oscillation. At the m-th travel pole the null vector R has exactly $m-1$ sign changes, for $m = 1, \dots, 5$ observed to be $0, 1, 2, 3, 4$ with the last change at $s = -, 3, 27, 75, 149$, so R is the m-th eigenvector of the travel operator and carries $m-1$ nodes. The portion beyond the last node dominates: the partial sums split as head $\approx -\Sigma_0(q_m)$ and tail $\approx 2\Sigma_0(q_m)$, so both weighted sums $\langle \lambda, R\rangle$ and $\langle R, \mu/e\rangle$ inherit the sign of the tail, which is the sign of $\Sigma_0(q_m)$. That is why the two factors share a sign and the product is positive. Making this a proof requires a quantitative tail-domination estimate uniform in m, which we do not have. A cheaper route is unavailable: the weight vectors are not positive, μ_s taking negative values at $2, 30, 33$ and 130 of its entries for $m = 2, 3, 4, 5$, so no bare positivity argument applies.

Fourth, this reduces $(R{-}J)$ to two explicit inequalities. Using $\sum_s q^s R_s = \Sigma_1(q_m) = 1$, the source's own decomposition gives

$$\langle \lambda, R\rangle = x\,B + C_\lambda, \qquad C_\lambda = \sum_\sigma \frac{B_\sigma}{2}\big(\Delta_\sigma^+ + \Delta_\sigma^-\big), \qquad \Delta_\sigma^\pm = \sum_{2s+1<\sigma\mp 1} \big(x^{\sigma\mp 1} - xq^s\big) R_s,$$

and we verified this identity numerically against the directly computed pairing at the first four poles. Every weight in C_λ is negative, since $2s+1 < \sigma \mp 1$ forces $x^{\sigma\mp 1} < x^{2s+1}$, so C_λ is a negatively weighted sum over the *head* of R, the part before its last node. Writing $\varepsilon_m = \operatorname{sign}\Sigma_0(q_m)$, the observed picture is that the head of R carries sign $-\varepsilon_m$, hence C_λ carries sign ε_m, and $|C_\lambda| > xB$, so $\langle \lambda, R\rangle$ carries sign ε_m; the same for $\langle L, \mu\rangle$, and the product is positive. Consequently $(R{-}J)$ follows from the two statements

(S1) the head of R carries sign $-\varepsilon_m$, for infinitely many m;

(S2) $|C_\lambda| > xB$ and the corresponding inequality for μ, for infinitely many m.

Both are inequalities about explicitly given finite sums, and neither is proved here. This is where we leave the problem: not at "show the pairing is non-zero" but at (S1) and (S2).

Both hold at every pole computed, and with sharp laws rather than marginally. For (S1) the ratio head$/(-\Sigma_0(q_m))$ takes the values 0.99634, 1.000475, 0.99999, 1.000044, 0.9999991, 1.000012, 1.000006, 1.00000 at $m = 2, \dots, 12$, so not merely the sign but the value is pinned: head $= -\Sigma_0(q_m)$ asymptotically, to seven digits by $m = 6$, equivalently the part of R beyond its last node sums to $2\Sigma_0(q_m)$. For (S2) the ratio $|C_\lambda|/(xB)$ takes the values 2.813, 3.352, 5.970, 6.533, 9.120, 9.687, 12.266, 12.835, 15.410, 15.980, 18.554 over the same range, increasing by alternately about 2.58 and 0.57, hence by about π every two steps, so $|C_\lambda|/(xB) \sim (\pi/2)m$ and the margin grows linearly in m.

At $m = 1$ both fail, head being empty and $|C_\lambda|/(xB) = 0.3995$, which is precisely the pole where positivity is available and $\Pi_y(q_1) > 0$ is already proved. The two regimes are therefore complementary: the dominant pole by Pringsheim positivity, and every computed pole above it by (S1) and (S2). What remains is to prove the two asymptotic laws, and it is worth saying exactly where that lands. The travel null vector is generated by $A_{s+1} - A_s = 2q(q^{2s}A_s + q^s B_s)$ and

$B_{s+1} - B_s = -2q(q^{3s}A_s + q^{2s}B_s)$ with $A_0 = 0$, $B_0 = 1$. Writing $\tau = -\log q$, $t = \tau s$ and $\varepsilon = \tau/2$, the continuum limit is the singularly perturbed system $\varepsilon Y' = M(t)Y$ with

$$M(t) = \begin{pmatrix} e^{-2t} & e^{-t} \\ -e^{-3t} & -e^{-2t} \end{pmatrix}, \qquad \operatorname{tr} M = 0, \quad \det M = 0, \quad M^2 = 0.$$

So M is nilpotent for every t, both eigenvalues vanishing identically. That is not an obstruction here; it is what makes the system solvable. Substituting $u = e^{-t}$ turns $\varepsilon Y' = MY$ into $\varepsilon Y' = -P(u)Y$ with $P = \left(\begin{smallmatrix} u & 1 \\ -u^2 & -u \end{smallmatrix}\right)$, so with $Y = (A, B)$ one has $\varepsilon A' = -(uA + B)$ and $\varepsilon B' = u(uA + B) = -u\,\varepsilon A'$. Eliminating B leaves

$$\varepsilon A'' + A = 0, \qquad \varepsilon = \tau/2, \qquad w := \varepsilon^{-1/2} = \sqrt{2/\tau}. \tag{3}$$

The boundary conditions are $A(1) = A_0 = 0$ and, from $B(0) = B_\infty = 0$, $A'(0) = 0$. Hence $A(u) = C\cos(wu)$, and $A(1) = 0$ forces the quantisation $\cos w = 0$, that is $w_m = (m - \frac{1}{2})\pi$, which is the recorded law $w_m \sim m\pi$; the relative error against $\sqrt{2/\tau_m}$ at the computed poles is $6.8 \cdot 10^{-3},\ 2.3 \cdot 10^{-3},\ 8.8 \cdot 10^{-4}, \ldots, 6.2 \cdot 10^{-5}$ for $m = 1, \ldots, 10$. Since $A(0) = \Sigma_0(q_m)$ we get $C = \Sigma_0(q_m)$ and

$$A(u) = \Sigma_0(q_m)\cos(wu), \qquad R_s \ \propto\ u\sin(wu), \qquad u = q^s.$$

This settles (S1) at leading order. The nodes of R sit at $u = k\pi/w$ with $1 \le k \le m-1$, so R has exactly $m - 1$ of them, which is the observed Sturm count at every computed pole; the last node in s is the smallest such u, namely π/w; and therefore head $= A(\pi/w) = \Sigma_0(q_m)\cos\pi = -\Sigma_0(q_m)$, which is exactly the observed law. Against the exact null vector, A_s matches $\Sigma_0\cos(wq^s)$ with maximal relative error $0.020,\ 0.011,\ 0.0069, \ldots, 0.0024$ for $m = 1, \ldots, 8$, decaying like $1/m$.

The bulk does *not* linearise the same way, and we record the failed attempt. Its recursion carries a source, and the same elimination formally gives $\varepsilon A'' + A = -\mathfrak{g}\beta$, the oscillator with constant forcing. That reduction does not survive checking. Imposing $A(1) = 0$, $A'(0) = -1/\varepsilon$ and the self-consistency $\beta = B(1)$ yields $\mathfrak{g}\beta = (-1)^m w$, which at the second pole predicts $+4.70$ against an actual $\mathfrak{g}\beta = -81.75$; and testing $\varepsilon A'' + A$ directly on the discrete bulk gives 80.1, 115.2, 113.1 at $u = 0.3, 0.5, 0.7$ rather than a constant. So the bulk boundary analysis above is wrong somewhere, and (S2) is not derived. The likely culprit is that $\mathfrak{g}_V = q/(1-q)$ grows like $w^2/2$, so the forcing is not uniformly small and a boundary layer near $u = 1$ is not negligible; we have not resolved it.

What is therefore established is (S1) at leading order, with the closed form verified above, and not (S2). Uniform control of the discrete-to-continuum error is also missing.

That last gap can be removed entirely, by not passing to a continuum at all. The telescoping noted below is not merely a structural remark: carried out, it solves the travel recursion in closed form.

Proposition 11.7 (the travel null vector in closed form). *Let A_s, B_s solve $A_{s+1} - A_s = 2q(q^{2s}A_s + q^sB_s)$ and $B_{s+1} - B_s = -2q(q^{3s}A_s + q^{2s}B_s)$ with $A_0 = 0$, $B_0 = 1$, and let $A_\infty = \lim_s A_s$, $B_\infty = \lim_s B_s$. Then, with $(q;q)_r = \prod_{i=1}^r (1 - q^i)$,*

$$A_\infty = \sum_{k \ge 1} (-1)^{k-1} \frac{2^k (1-q)^{k-1} q^{k + (k-1)^2}}{(q;q)_{2k-1}}, \qquad B_\infty = \sum_{k \ge 0} \frac{2^k (q-1)^k q^{k^2}}{(q;q)_{2k}}.$$

Proof. Write the transfer as $M_s = I + c_sN_s$ with $c_s = 2q^{2s+1}$ and $N_s = v_sw_s^\top$, where $v_s = (q^{-s/2}, -q^{s/2})^\top$ and $w_s = (q^{s/2}, q^{-s/2})^\top$; then $w_s^\top v_s = 0$, so $N_s^2 = 0$. Since each factor is a rank-one update, the ordered product expands exactly over increasing chains,

$$\prod_s (I + c_s v_s w_s^\top) = I + \sum_{k\ge 1} \sum_{s_1<\cdots<s_k} \Big(\prod_i c_{s_i}\Big)\Big(\prod_{i=2}^{k} w_{s_i}^\top v_{s_{i-1}}\Big) v_{s_k} w_{s_1}^\top,$$

with link factors $w_a^\top v_b = q^{(a-b)/2} - q^{(b-a)/2}$. Applying this to $(0,1)^\top$ and writing $\delta_i = s_i - s_{i-1} \ge 1$, the half-integer powers cancel: the A component acquires weight $q^{2\sum s_i + k - s_k}$ and the B component $q^{2\sum s_i + k}$, each carrying $\prod_{i\ge 2}(1-q^{\delta_i})$ and a sign. Substituting $\sum s_i = ks_1 + \sum_{i\ge 2}(k-i+1)\delta_i$ makes s_1 and the δ_i independent, so every sum is geometric. For B the s_1 sum gives $1/(1-q^{2k})$ and the δ_i sum gives, with $j = k-i+1$,

$$\sum_{\delta\ge 1} q^{2j\delta}(1-q^\delta) = \frac{q^{2j}}{1-q^{2j}} - \frac{q^{2j+1}}{1-q^{2j+1}} = \frac{q^{2j}(1-q)}{(1-q^{2j})(1-q^{2j+1})}.$$

Multiplying over $j = 1,\dots,k-1$ contributes $q^{k^2-k}(1-q)^{k-1}\cdot(1-q)/(q;q)_{2k-1}$, and combining with q^k and $1/(1-q^{2k})$ gives the stated B_∞. The computation for A is identical with $2j$ replaced by $2j-1$, where the same cancellation gives $q^{2j-1}(1-q)/\big((1-q^{2j-1})(1-q^{2j})\big)$, and the products telescope to $q^{(k-1)^2}(1-q)^{k-1}/(q;q)_{2k-2}$; the s_1 sum contributes $1/(1-q^{2k-1})$, and $(q;q)_{2k-2}(1-q^{2k-1}) = (q;q)_{2k-1}$. $\square$

Both identities were checked against the recursion in exact rational arithmetic at $q = \frac{1}{2}, \frac{1}{3}, \frac{2}{5}, \frac{1}{10}, \frac{3}{7}, \frac{1}{5}$, agreeing to every digit computed.

Corollary 11.8 (exact quantisation). *The travel poles are exactly the zeros in* $(0,1)$ *of the entire-in-*q *series* $B_\infty(q) = \sum_{k\ge 0} 2^k (q-1)^k q^{k^2}/(q;q)_{2k}$.

Numerically the first ten zeros are 0.44945363, 0.91348664, 0.96804175, 0.98357902, 0.99003740, 0.99332100, 0.99521395, 0.99640323, 0.99719876, 0.99775693, and the resulting $w = \sqrt{2/\tau}$ reproduces the relative errors against $(m - \frac{1}{2})\pi$ recorded above, namely $6.8\cdot 10^{-3}$, $2.3\cdot 10^{-3}$, $8.8\cdot 10^{-4}$ and so on.

Remark 11.9 (what this does and does not close). Proposition 11.7 removes the discrete-to-continuum error from the quantisation step: the poles are not approximated by $\cos w = 0$, they are the zeros of an explicit series, so no Liouville–Green estimate is needed to locate them and the uniformity gap noted above is gone at that step. It does not by itself prove the head and tail laws that (S1) asserts, which are statements about the profile of A_s at intermediate s rather than about A_∞, and it says nothing about (S2). What it changes is the kind of estimate required: the remaining questions are now about explicit q-series with q at a zero of another explicit q-series, rather than about the fidelity of a continuum approximation.

There is, however, a reason to expect the continuum route to be the wrong one, and it is structural. Both chains are exactly solvable as discrete transfer products, and passing to a continuum destroys that. Writing the travel recursion as $(A_{s+1}, B_{s+1})^\top = M_s(A_s, B_s)^\top$ gives

$$M_s = \begin{pmatrix} 1+2q^{1+2s} & 2q^{1+s} \\ -2q^{1+3s} & 1-2q^{1+2s} \end{pmatrix}, \qquad \det M_s = 1,$$

so the transfer lies in SL_2. Moreover $M_s = I + c_s N_s$ with $c_s = 2q(q^2)^s$ and $N_s^2 = 0$, and $N_s = D_s^{-1} N D_s$ for the *fixed* nilpotent $N = \left(\begin{smallmatrix} 1 & 1 \\ -1 & -1 \end{smallmatrix}\right)$ and $D_s = \mathrm{diag}(q^{s/2}, q^{-s/2})$, while $D_s D_{s+1}^{-1} = \mathrm{diag}(q^{-1/2}, q^{1/2})$ is constant. The ordered product therefore telescopes into an alternating product of a fixed unipotent family and a fixed diagonal, with purely geometric coefficients. The bulk chain has the same shape, $M_b = I + c_b N_b$ with $c_b = 2(q^2)^b$ and again $\det M_b = 1$, differing from the travel chain only in the prefactor $2q$ against 2 and in carrying a source.

Two consequences. First, geometric coefficients in SL_2 is precisely the q-hypergeometric setting in which the Hahn–Exton q-cosine lives, so the appearance of that function is structural rather than incidental. Second, because $\det M = 1$ the discrete Wronskian is conserved, so the inhomogeneous bulk chain is the homogeneous one plus a particular solution obtained by variation of parameters, with an exact discrete Green's function and no continuum approximation anywhere. That is the form in which (S2) should be attacked.

The two chains are moreover the same chain. Writing $q = r^2$ and taking $a = q^s$ for the travel argument,

$$D\, M^{\text{travel}}(r^2, a) = M^{\text{bulk}}(ar)\, D, \qquad D = \operatorname{diag}(1, r),$$

an identity in the polynomial ring, with no hypotheses, no inverse and no fourth root. It is VERIFIED in `lean/with_mathlib/TransferChain.lean` as `half_shift`, together with det = 1 for both chains and the nilpotence of their increments. An earlier form of this statement carried the hypotheses $r^2 = q$ and $r\,r^{-1} = 1$ and a conjugator involving $q^{1/4}$; none of that is needed, and the simplification was found in the course of formalising. This is the precise form of the statement that the zeros of S_e and of $1 - \Sigma_1$ are those of one q-cosine sampled a half q-step apart: the two chains differ by a half-step in the index and a constant conjugation, nothing else. In particular the homogeneous solutions of the bulk chain are those of the travel chain at shifted index, so the inhomogeneous bulk problem is closed by the conserved Wronskian alone.

The homogeneous bulk chain can now be solved outright, because the telescoping of Proposition 11.7 used only the rank-one nilpotent structure, which the bulk shares.

Proposition 11.10 (the homogeneous bulk chain in closed form). *Let A_b, B_b solve $A_{b+1} - A_b = 2(q^{2b}A_b + q^b B_b)$ and $B_{b+1} - B_b = -2(q^{3b}A_b + q^{2b}B_b)$ with $A_0 = 0$, $B_0 = 1$. Then*

$$A_\infty^{\text{bulk}} = \sum_{k\ge1} (-1)^{k-1} \frac{2^k (1-q)^{k-1} q^{(k-1)^2}}{(q;q)_{2k-1}}, \qquad B_\infty^{\text{bulk}} = \sum_{k\ge0} \frac{2^k (q-1)^k q^{k(k-1)}}{(q;q)_{2k}}.$$

Proof. Identical to Proposition 11.7. The only change is that $\prod_i c_{b_i} = 2^k q^{2\sum b_i}$ rather than $2^k q^{2\sum s_i + k}$, since $c_b = 2(q^2)^b$ carries no extra factor q; every exponent therefore loses q^k, and the geometric sums are unchanged. □

Both identities were checked against the recursion in exact rational arithmetic at $q = \frac{1}{2}, \frac{1}{3}, \frac{2}{5}, \frac{3}{7}, \frac{1}{10}, \frac{1}{5}$.

Corollary 11.11 (the two chains are one q-cosine at two arguments). *With* $\Phi(z) = \sum_{n\ge0} (-1)^n q^{n(n-1)} z^n / \big((q^2;q^2)_n(q$ ${}_1\phi_1(0; q; q^2, z)$ *one has*

$$B_\infty = \Phi\big(2q(1-q)\big), \qquad B_\infty^{\text{bulk}} = \Phi\big(2(1-q)\big).$$

Proof. $(q^2;q^2)_n (q;q^2)_n = (q;q)_{2n}$, and $q^{n(n-1)} z^n$ with $z = 2q(1-q)$ gives $2^n (q-1)^n q^{n^2}$, while $z = 2(1-q)$ gives $2^n (q-1)^n q^{n(n-1)}$. □

This is the precise content of the half-step statement made above: the two arguments differ by the single factor q. It also removes the last continuum approximation from the homogeneous part of the problem. What (S2) still needs is the inhomogeneous bulk, that is the particular solution against the source, and then an inequality; by the conserved Wronskian the particular solution is variation of parameters against the two explicit homogeneous solutions above, so the remaining work is an estimate on an explicit discrete Green's function rather than on a continuum approximation. We have not carried it out.

Consistently with this, the bulk partial sums do fit the same oscillatory closed form. A least squares fit of A_b to $C_1 \cos(Wu) + C_2 \sin(Wu) + K$ with $u = q^b$ has relative residual $1.3 \cdot 10^{-7},\ 8.2 \cdot 10^{-4},\ 1.9{\cdot}10^{-3},\ 2.0{\cdot}10^{-3},\ 5.0{\cdot}10^{-4}$ at the first five poles, with $W/\sqrt{2/\tau} = 1.2,\ 1.05,\ 1.015,\ 1.01,\ 1.005$. The bulk is an oscillator with the same limiting frequency; the earlier failures came from imposing the travel frequency and the travel boundary data on it, not from any absence of the closed form.

One gap in the above should be recorded rather than passed over. Deriving the travel equation from $\mathrm{d}A/\mathrm{d}s = 2q^{1+s}(q^s A + B)$ and $\mathrm{d}u/\mathrm{d}s = -\tau u$ gives $\varepsilon = \tau/2q$ and hence $w = \sqrt{2q/\tau}$, whereas the quantisation is matched by $\sqrt{2/\tau}$, and by a margin: at the tenth pole the two candidates miss $(m - \frac{1}{2})\pi$ by $1.2 \cdot 10^{-3}$ and $6.2 \cdot 10^{-5}$ respectively. A factor of q is unaccounted for, and until it is explained the closed form of (3) is supported by its verification and not by its derivation.

We record the exact structure, and not any consequence for (S2), which we have not derived.

Fifth, a Blaschke argument cannot be used, and this is worth recording so that it is not attempted. If Π_y vanished at all but finitely many q_m, one would like to conclude that the numerator vanishes identically. That needs the zeros to violate the Blaschke condition. They do not: writing $\tau_m = -\log q_m$, the products $m^2\tau_m$ are 0.800, 0.362, 0.292, 0.265, 0.250, 0.241, 0.235, 0.231, 0.227, 0.225, 0.222, 0. for $m = 1, \ldots, 14$, so $1-q_m \asymp m^{-2}$ and $\sum_m (1-q_m)$ converges, to about 0.72. A function of bounded characteristic on the disc may therefore vanish at every q_m without vanishing identically, and no contradiction follows.

Remark 11.12 (the arithmetic structure at $(1,2,3)$). For the generic form of Problem 11.13 a single faithful tuple suffices, and by Corollary 4.4 it is enough to find one at which the *point group* is faithful. The tuple $(1,2,3)$, where the point group agrees with the envelope through depth 22, carries an arithmetic structure that the others do not, and we record it because it is what any attempt should use.

The two denominators are $|m_1|^2 = a_1^2 + a_2^2 = 5$ and $|m_2|^2 = a_2^2 + a_3^2 = 13$, both prime. So R_0, R_2, R_3 lie in $GL_3(\mathbb{Z}_5)$ while $v_5(R_1) = -1$, and R_0, R_1, R_3 lie in $GL_3(\mathbb{Z}_{13})$ while $v_{13}(R_2) = -1$. The generators are orthogonal for the standard form, and $x^2 + y^2 + z^2$ is isotropic over $\mathbb{Q}_5$ and over $\mathbb{Q}_{13}$, so SO_3 is split at both places and its Bruhat–Tits building there is a *tree* rather than a rank-two building. Finally the graph splits: with $\Gamma_1 = \{0,2,3\}$ and $\Gamma_2 = \{0,1,3\}$ one has $\Gamma_1 \cap \Gamma_2 = \{0,3\}$, and 1 and 2 are non-adjacent in Γ_3, so

$$W_3 \;=\; (\mathbb{Z}/2 \times D_\infty) \;*_{\,\mathbb{Z}/2\times\mathbb{Z}/2}\; (D_\infty \times \mathbb{Z}/2),$$

the first factor $\langle R_0, R_2, R_3\rangle$ being 5-integral and the second $\langle R_0, R_1, R_3\rangle$ being 13-integral, with the amalgamated $\langle R_0, R_3\rangle$ integral at both. Both factors are faithfully represented: each contains a D_∞ acting as rotations by an angle with rational cosine $3/5$ respectively $-5/13$, which has infinite order by Niven's theorem, and in each the extra involution reverses an axis that the D_∞ fixes. What is missing is the step from these data to injectivity of the amalgam.

The obstacle is that this is a ping-pong on *two* trees, one at each place, and a group acting on a product of trees is outside Bass–Serre theory. That can be removed. Solving $a_1^2 + a_2^2 = 5^j$ and $a_2^2 + a_3^2 = 5^k$ in distinct positive integers gives $(1,2,11)$, where $|m_1|^2 = 5$ and $|m_2|^2 = 125$, so the whole point group lies in $O_3(\mathbb{Z}[1/5])$ and acts on the *single* tree at $p = 5$, with R_0, R_3 integral and $v_5(R_1) = -1$, $v_5(R_2) = -3$. The point group there agrees with the envelope through depth 21 at both primes. So $(1,2,11)$, not $(1,2,3)$, is the tuple at which the argument should be attempted: one tree, and Bass–Serre theory applies to it. We have not carried the argument out, and record only that the arithmetic preconditions hold.

Problem 11.13 (The alternative in dimension $n \geq 4$)**.** Is $N_{\text{gen}} = \bigcap_a \ker \rho_a$ trivial, equivalently is ρ_a injective for generic a? A single faithful tuple answers this affirmatively. The arithmetic form is FALSE (Theorem 4.2), which does not decide the generic form.

Remark 11.14 (Class C faithfulness)*.* The conjecture that ρ_a is injective for every $a \in \mathbb{Q}^n_{>0}$ with $e(a) = t(a) = 0$ and $n \geq 3$ is FALSE, by Theorem 4.2. Only the generic form (a) survives.

Problem 11.15 (Ambient embedding)**.** For $n \geq 3$, does W_n embed in $O(n)$? By Proposition 3.2 this is equivalent to the ambient embedding question for $\mathrm{Isom}(\mathbb{R}^n)$, and by Remark 3.5 it asks whether W_n has a faithful orthogonal representation of degree n.

Problem 11.16 (Finite presentation)**.** For $n \geq 3$, is W_n/N_{gen} finitely presented? A negative answer refutes Problem 11.13(a). At $n = 2$ the answer is no.

Problem 11.17 (Transcendence of the zeros)**.** Are the zeros $z_k(q)$ of the Hahn–Exton q-cosine transcendental at algebraic $q \in (0, 1)$, and is β_2 transcendental? This is the q-analogue of Siegel's theorem on the zeros of Bessel functions. Even irrationality of β_2 is open.

References

[1] E. B. Vinberg, *Discrete linear groups generated by reflections*, Izv. Akad. Nauk SSSR Ser. Mat. **35** (1971), 1072–1112; English translation, Math. USSR-Izv. **5** (1971), 1083–1119.

[2] P.-E. Caprace and K. Fujiwara, *Rank-one isometries of buildings and quasi-morphisms of Kac–Moody groups*, Geom. Funct. Anal. **19** (2010), 1296–1319.

[3] J. Danciger, F. Guéritaud and F. Kassel, *Proper affine actions for right-angled Coxeter groups*, Duke Math. J. **169** (2020), 2231–2280.

[4] D. Osin, *Acylindrically hyperbolic groups*, Trans. Amer. Math. Soc. **368** (2016), 851–888.